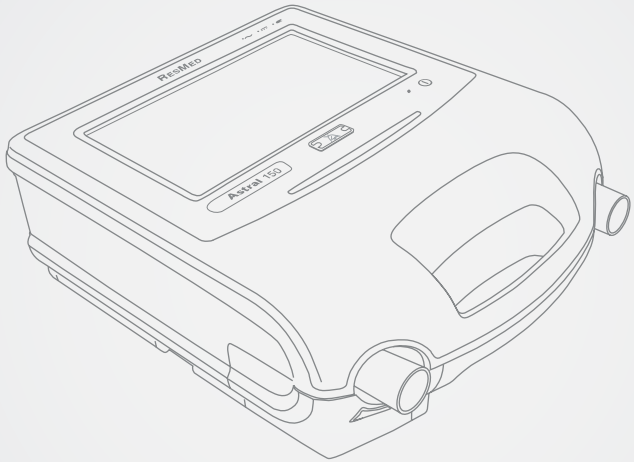
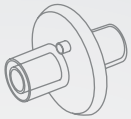
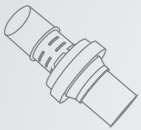
ResMed

Astral™ series

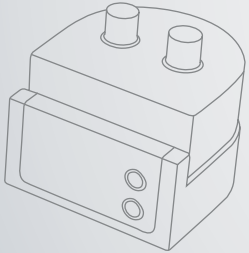
Life support ventilators



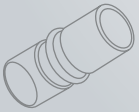












Astral best practice

A guide with precautionary setup options to reduce water in the patient circuit

ResMed.com/Astral

Introduction

- Water condensation is a known side effect of ventilation, especially when using a humidifier.
- As this is a multifactorial problem, no one solution is able to solve all related events.
- The purpose of "Astral best practice" is to explain and describe different options that are currently available to minimise condensation, particularly when using a humidifier.
- The brochure covers several options applied in practice by many Astral users around Europe. It is a complement to the user guides and is perfectly aligned with warnings and recommendations included in the regulatory documentation.
- For additional information, please refer to the Astral user guide and the instructions for any other devices that are included in the circuit.

Signs & symbols



Coaxial
double limb circuit



Single limb valve circuit,
inspiratory side
of double limb circuit



Expiratory side
of double limb circuit



Single circuit



Double limb circuit,
heated expiratory tube
with membrane



Antibacterial filter
(AB filter)



Mechanical / HEPA
filter



Combi filter
(HME/AB filter)



Heat and moisture
exchanger (HME)



invasive



non-invasive
mask



non-invasive
mouthpiece



heated setup
with humidifier

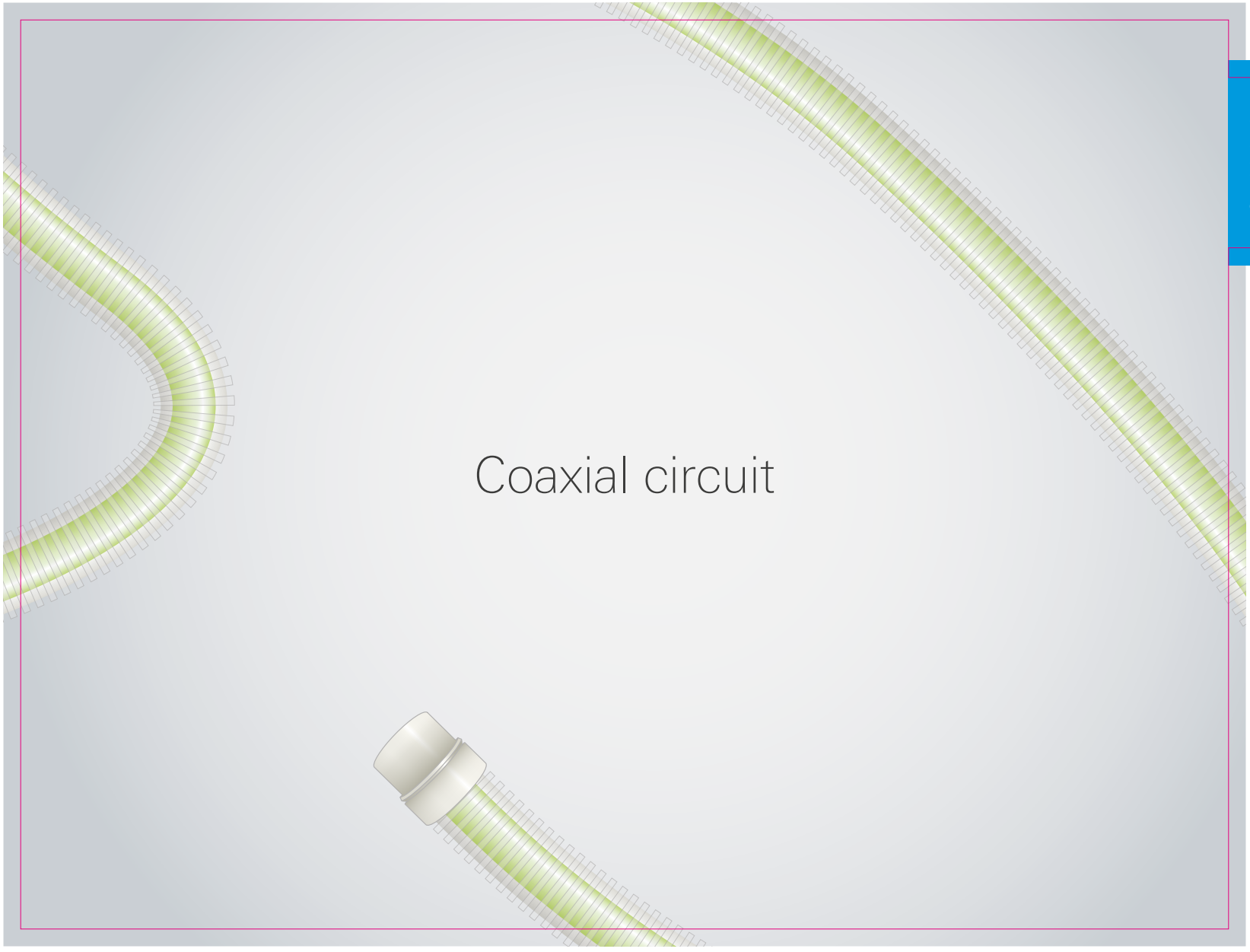


Aerogen™
nebuliser



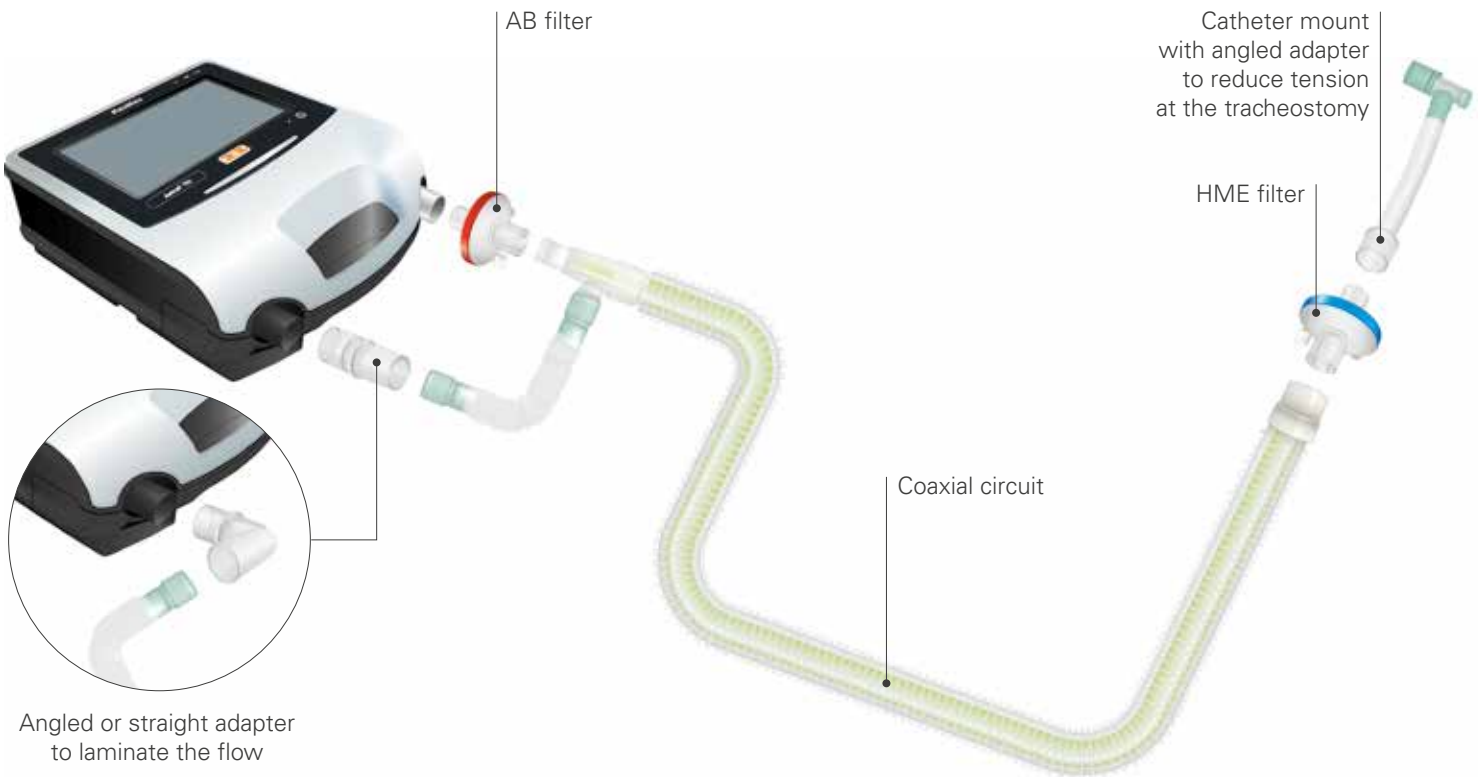
handy hint

Coaxial circuit

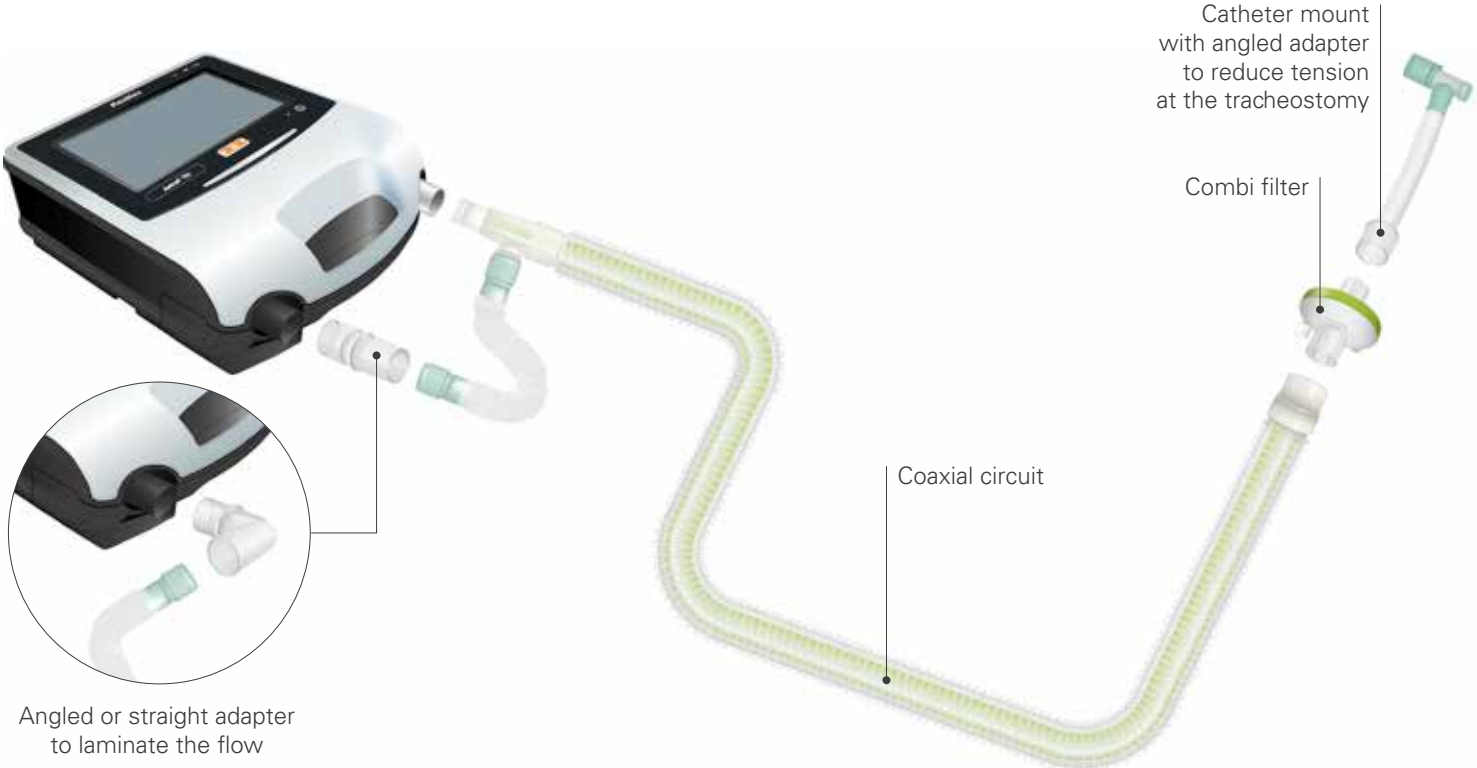


Straight or angled adapter in front of expiratory port

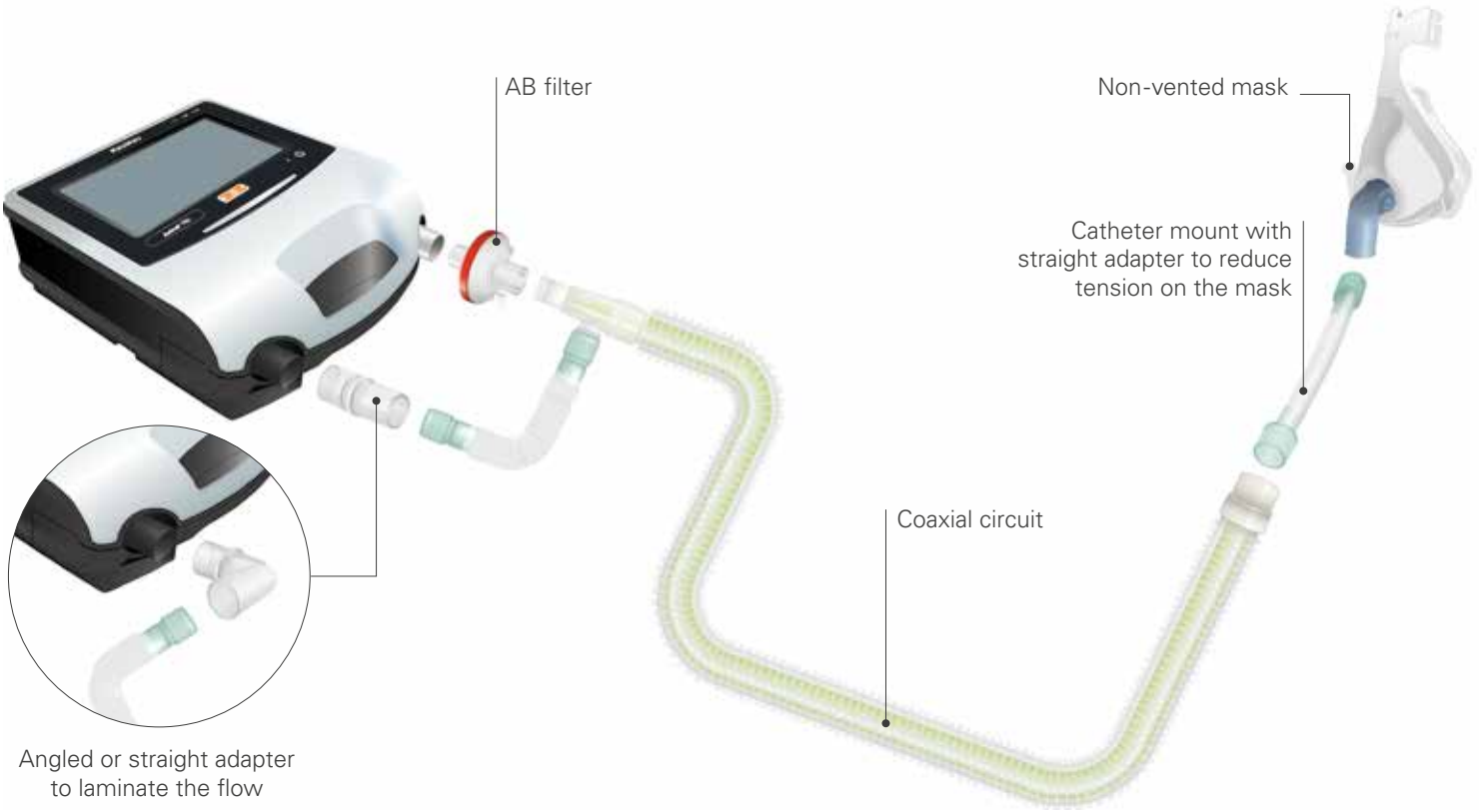
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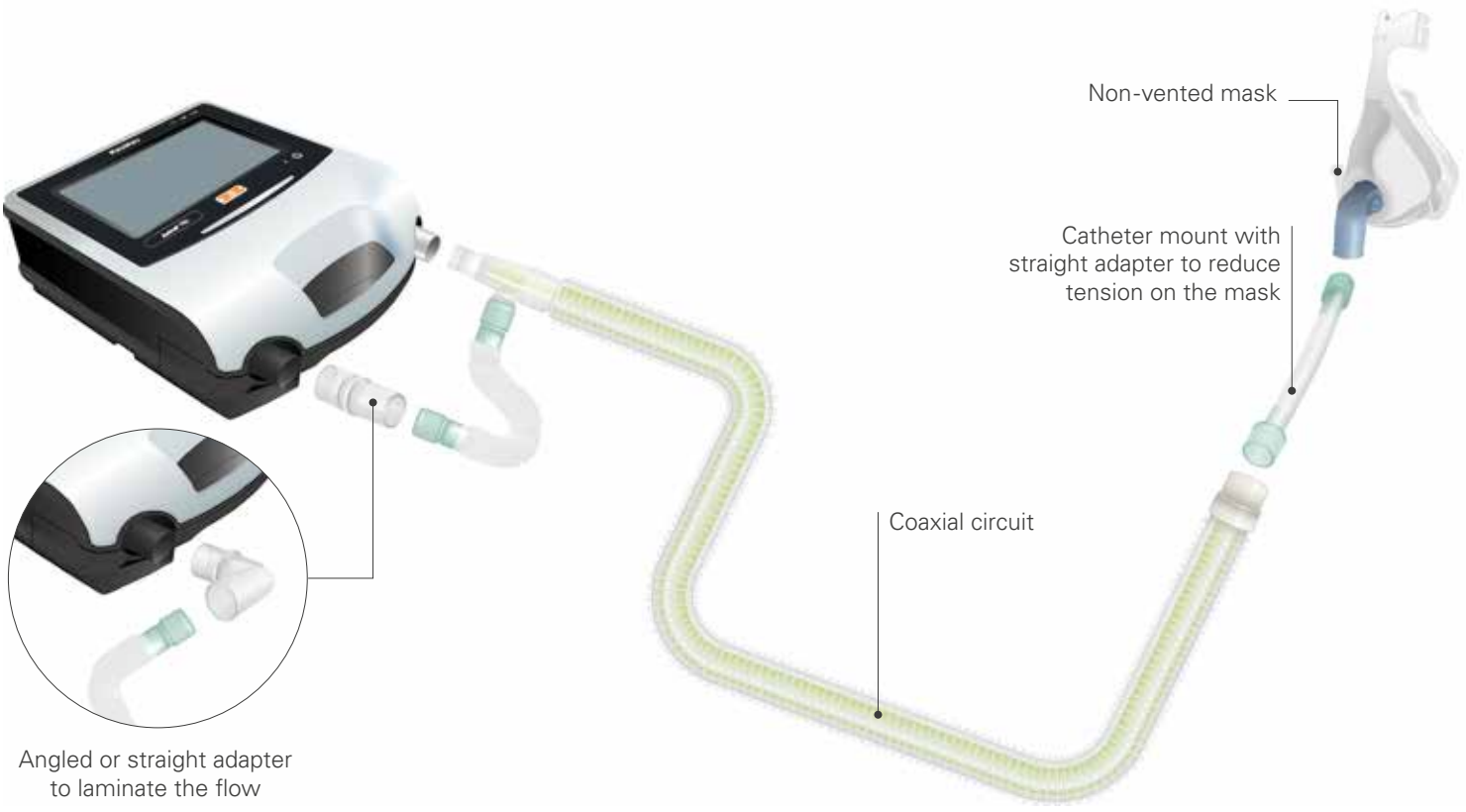
Straight or angled adapter in front of expiratory port



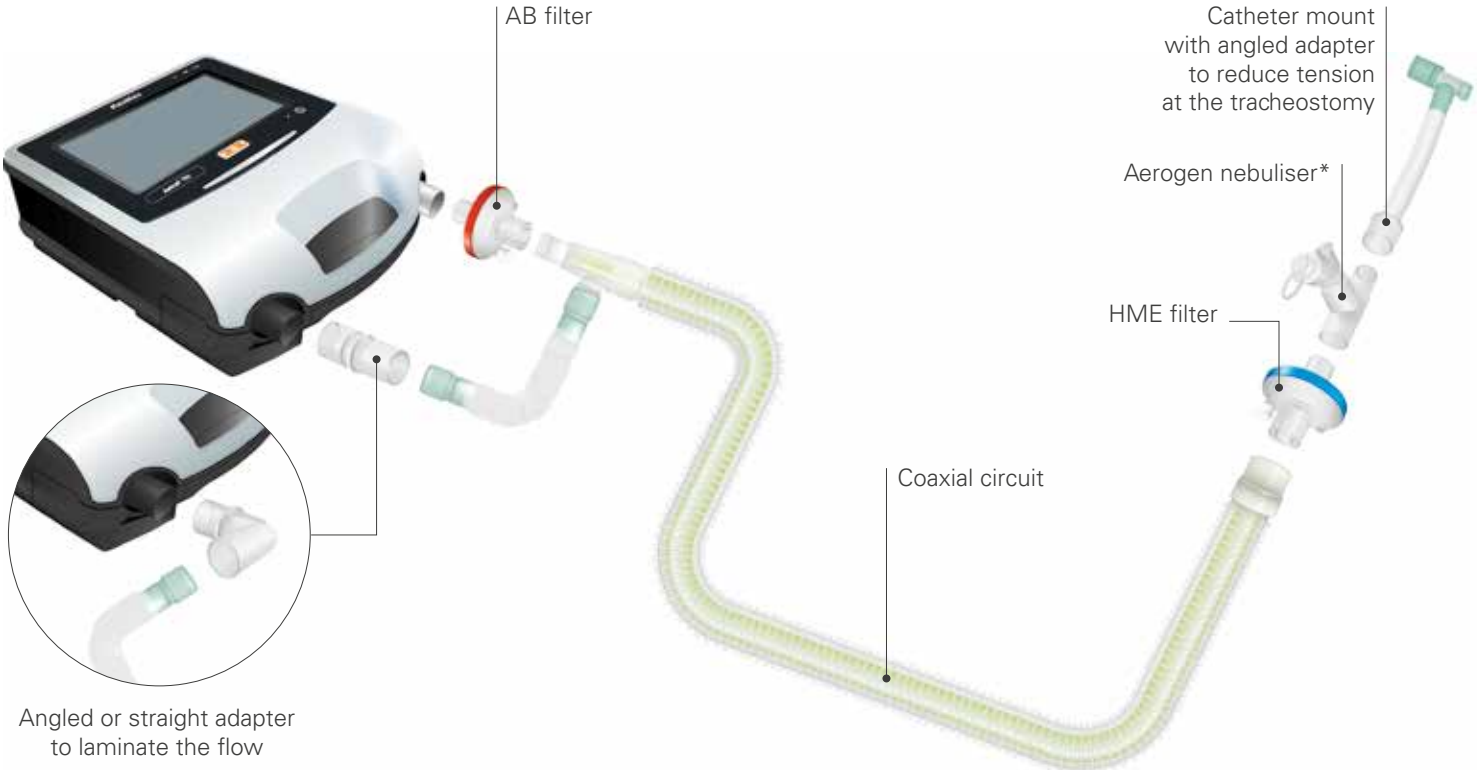
Straight or angled adapter in front of expiratory port



Straight or angled adapter in front of expiratory port

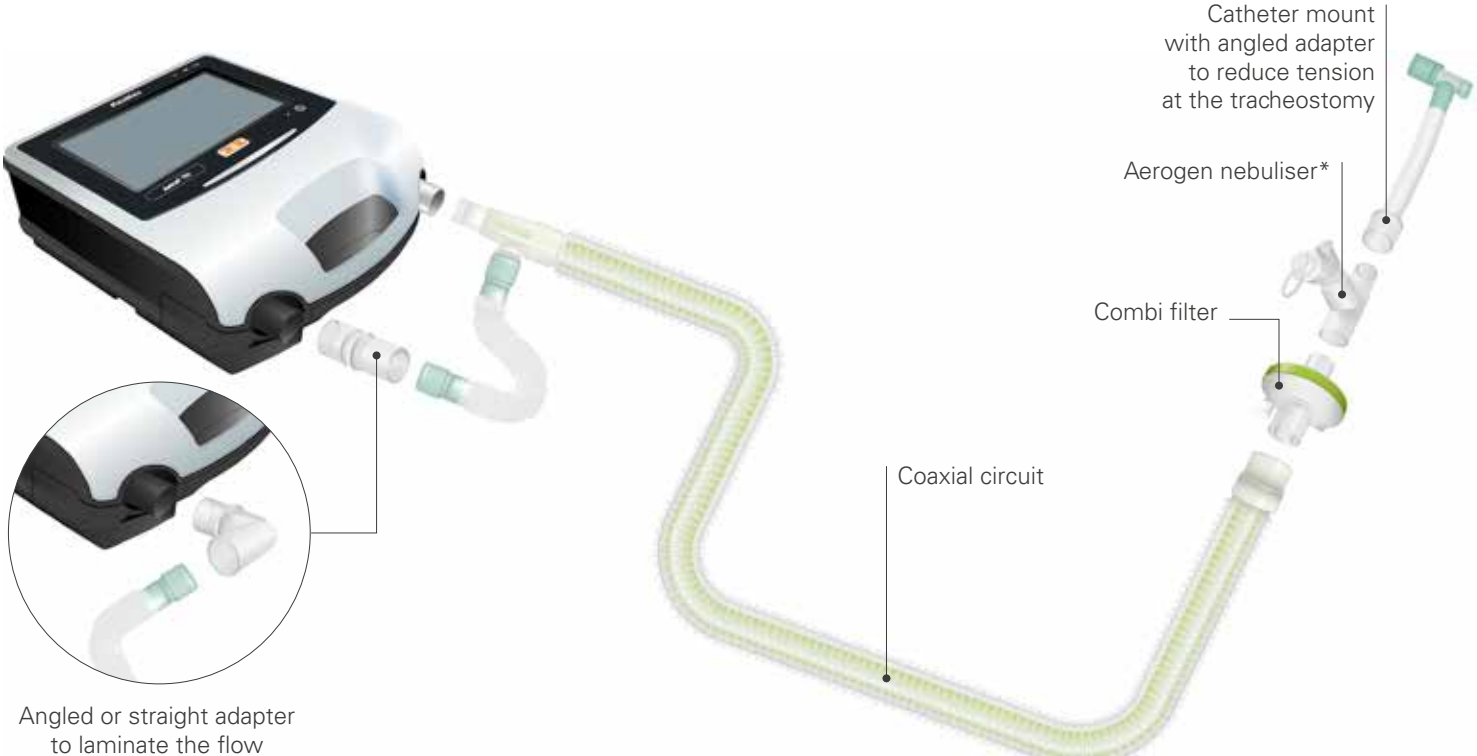


Straight or angled adapter in front of expiratory port



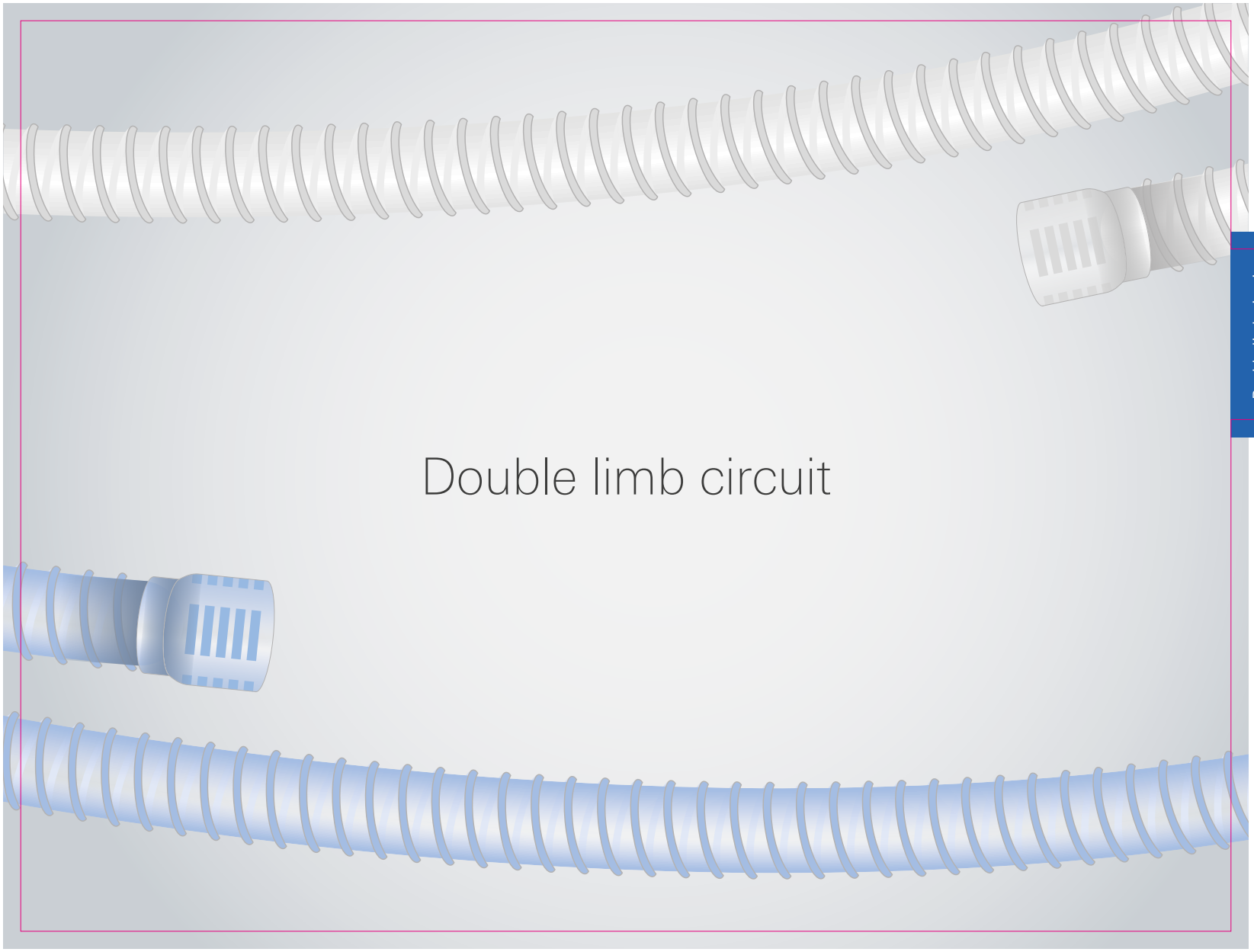
* Please see Aerogen user guide for details.

Straight or angled adapter in front of expiratory port



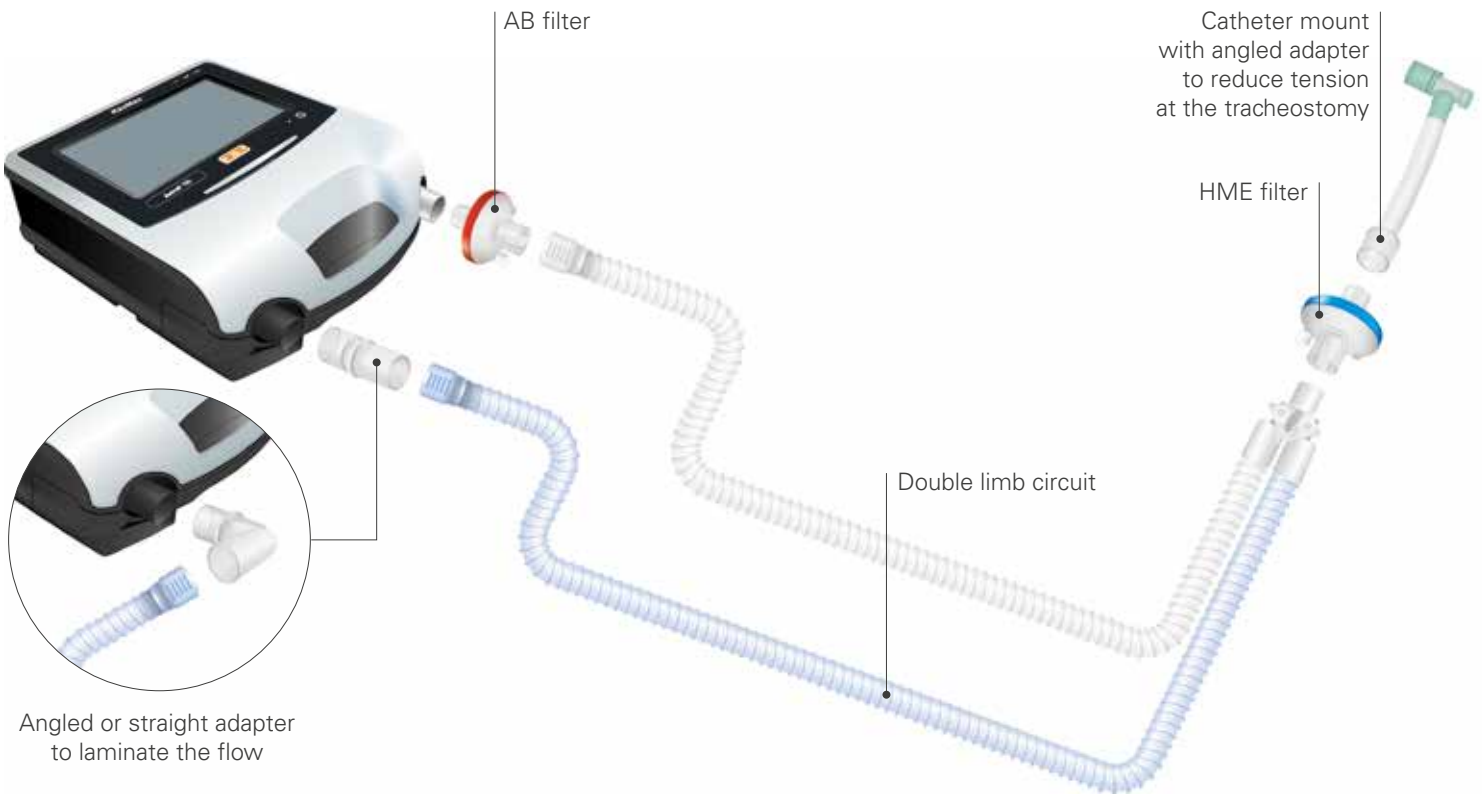
* Please see Aerogen user guide for details.

Double limb circuit

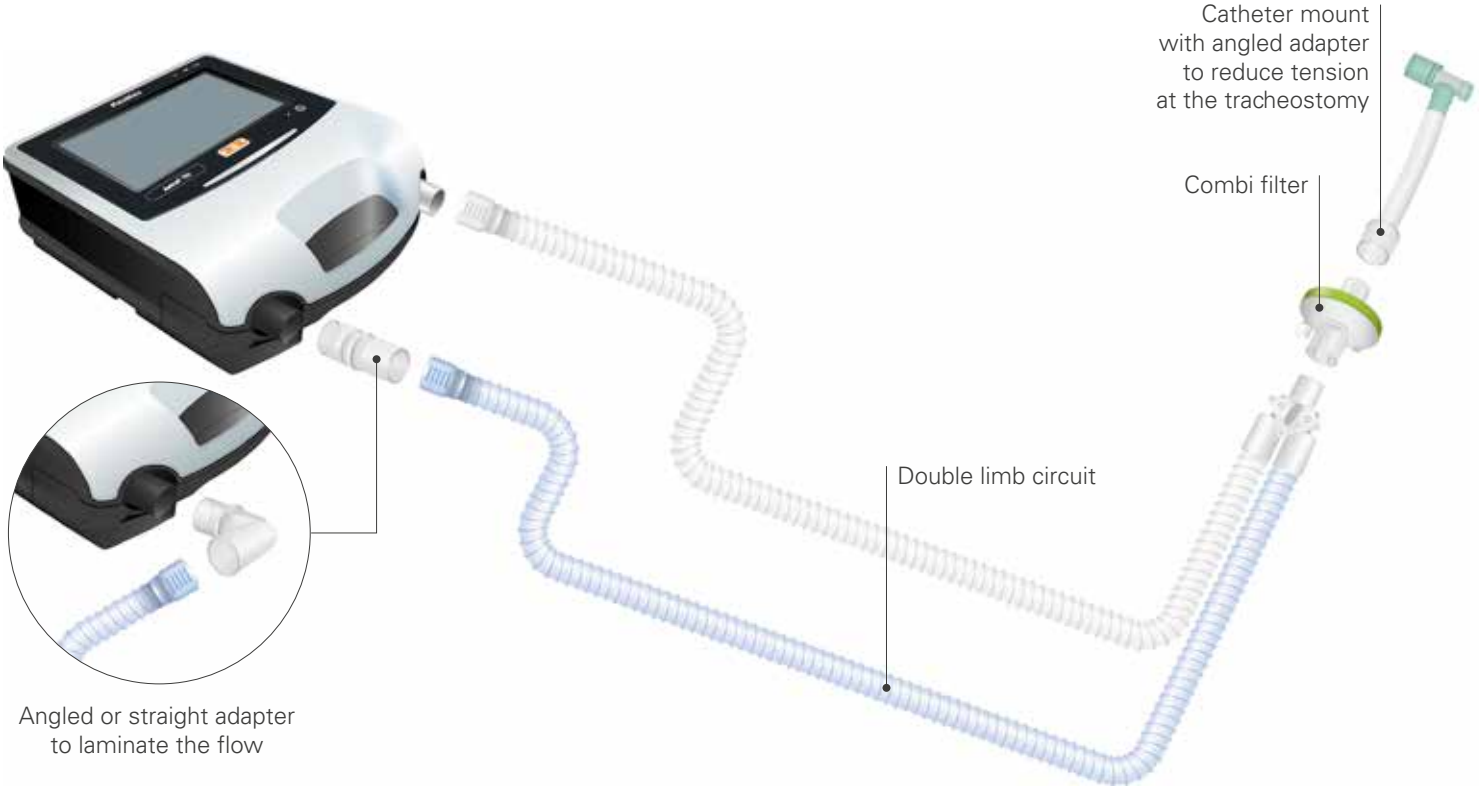


Straight or angled adapter in front of expiratory port

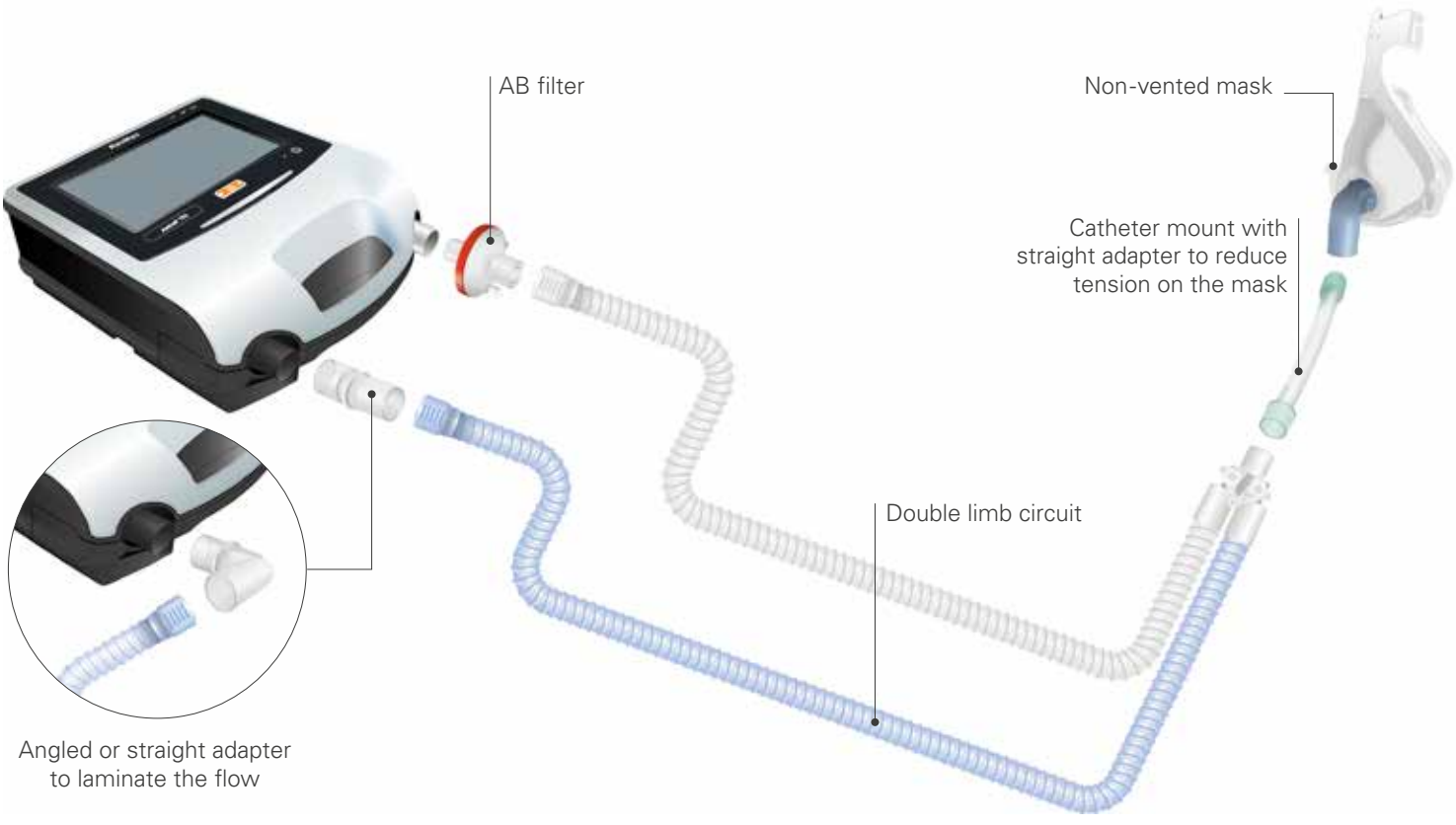
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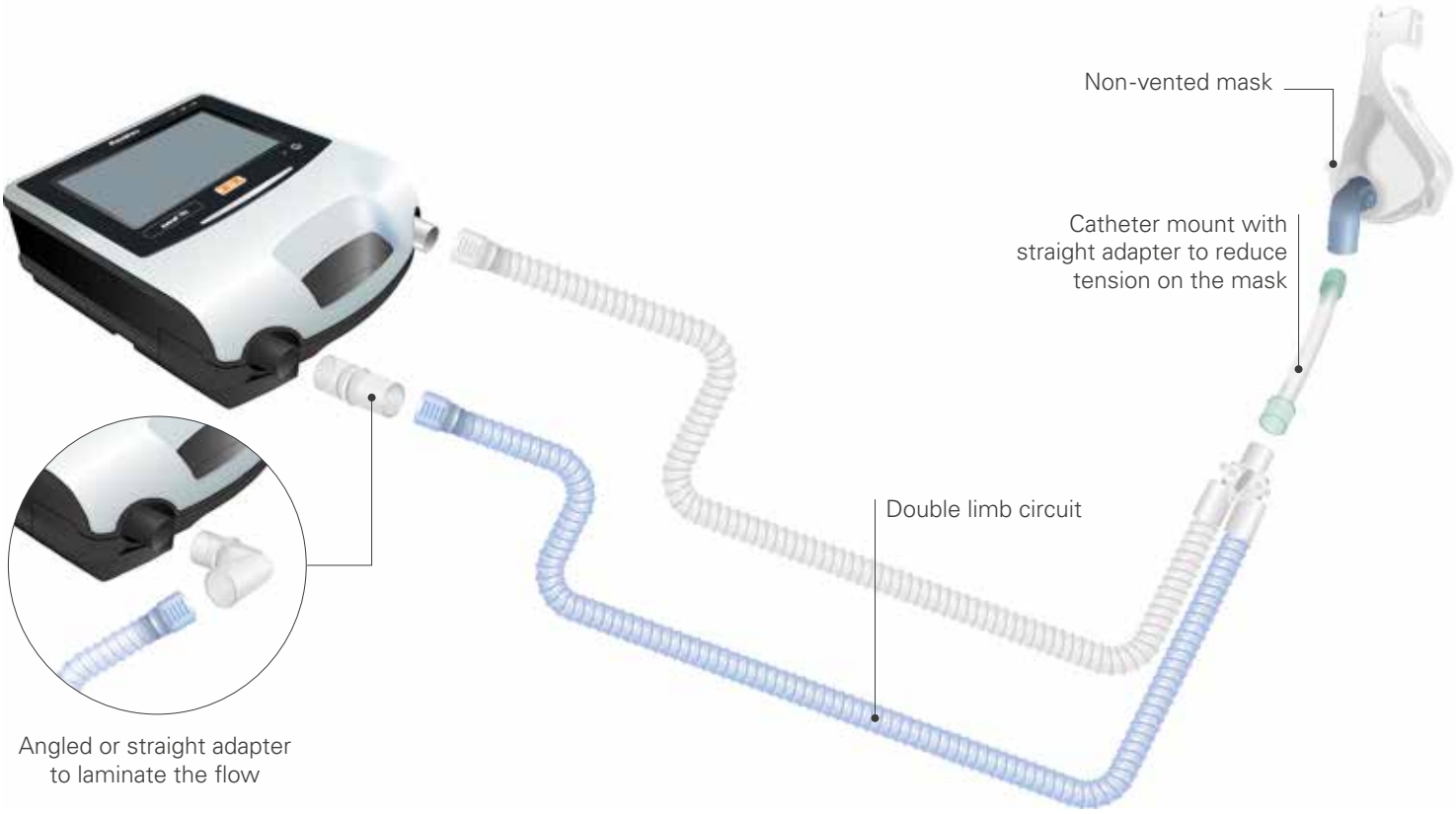
Straight or angled adapter in front of expiratory port

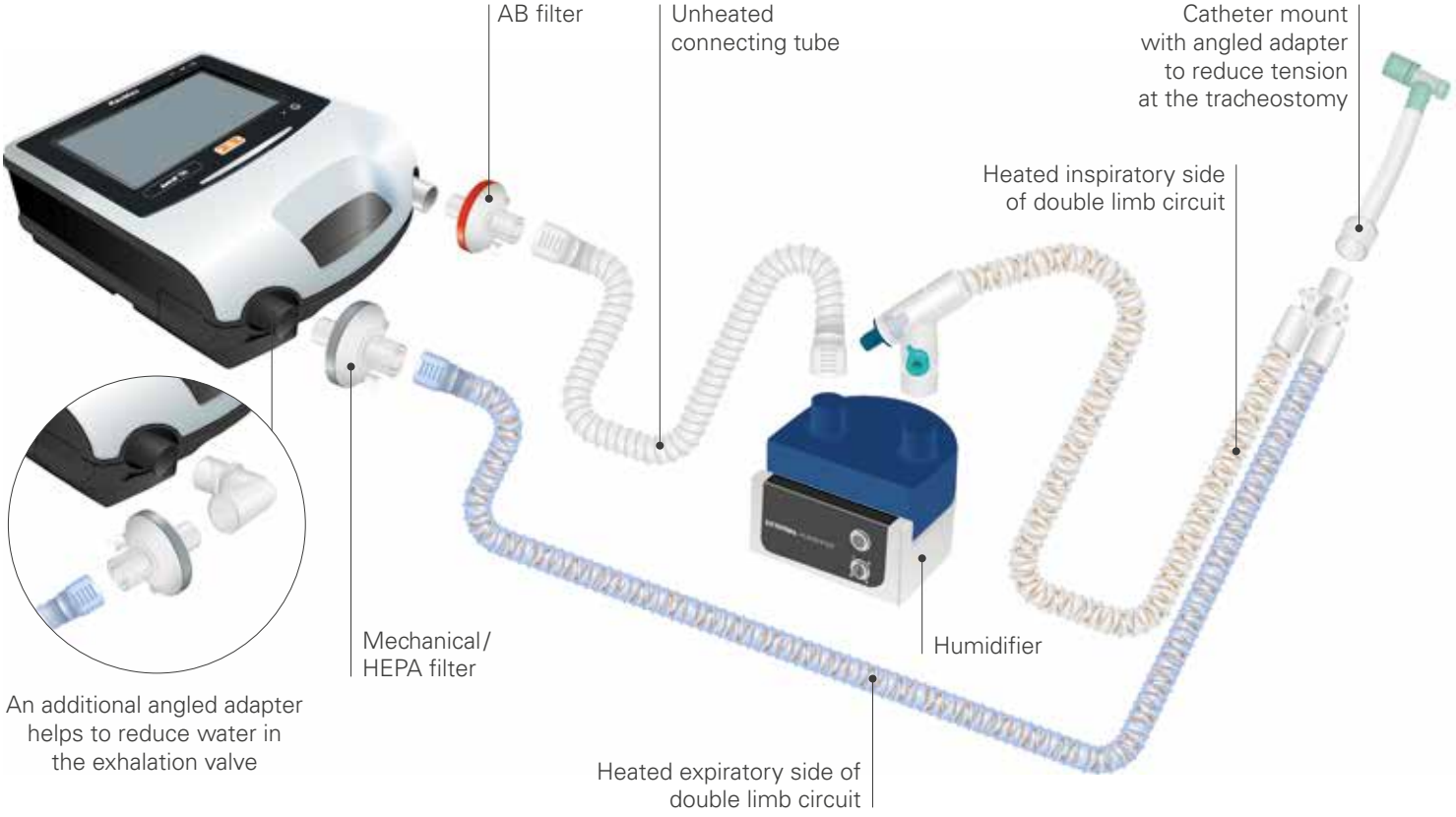


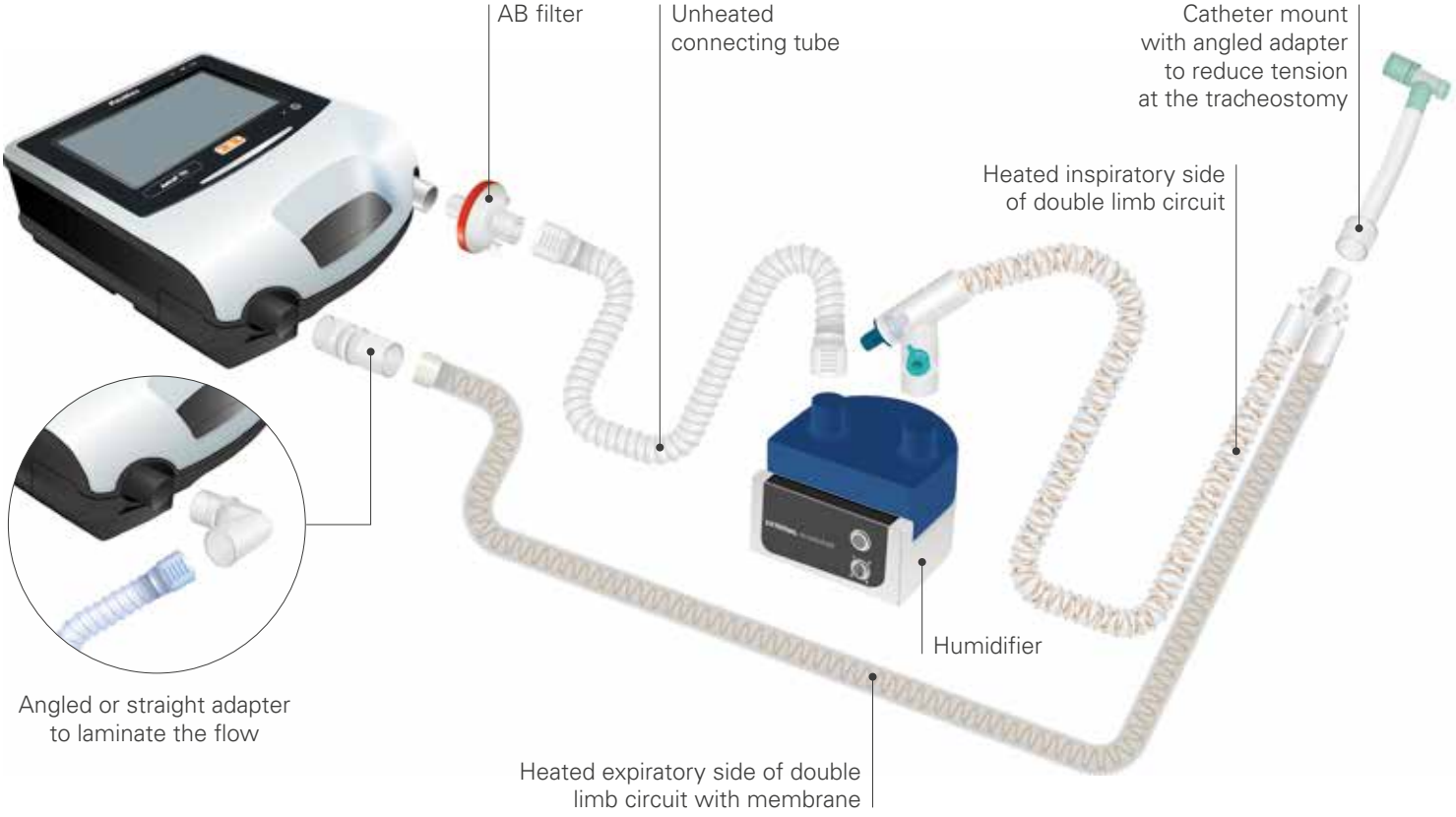
Straight or angled adapter in front of expiratory port

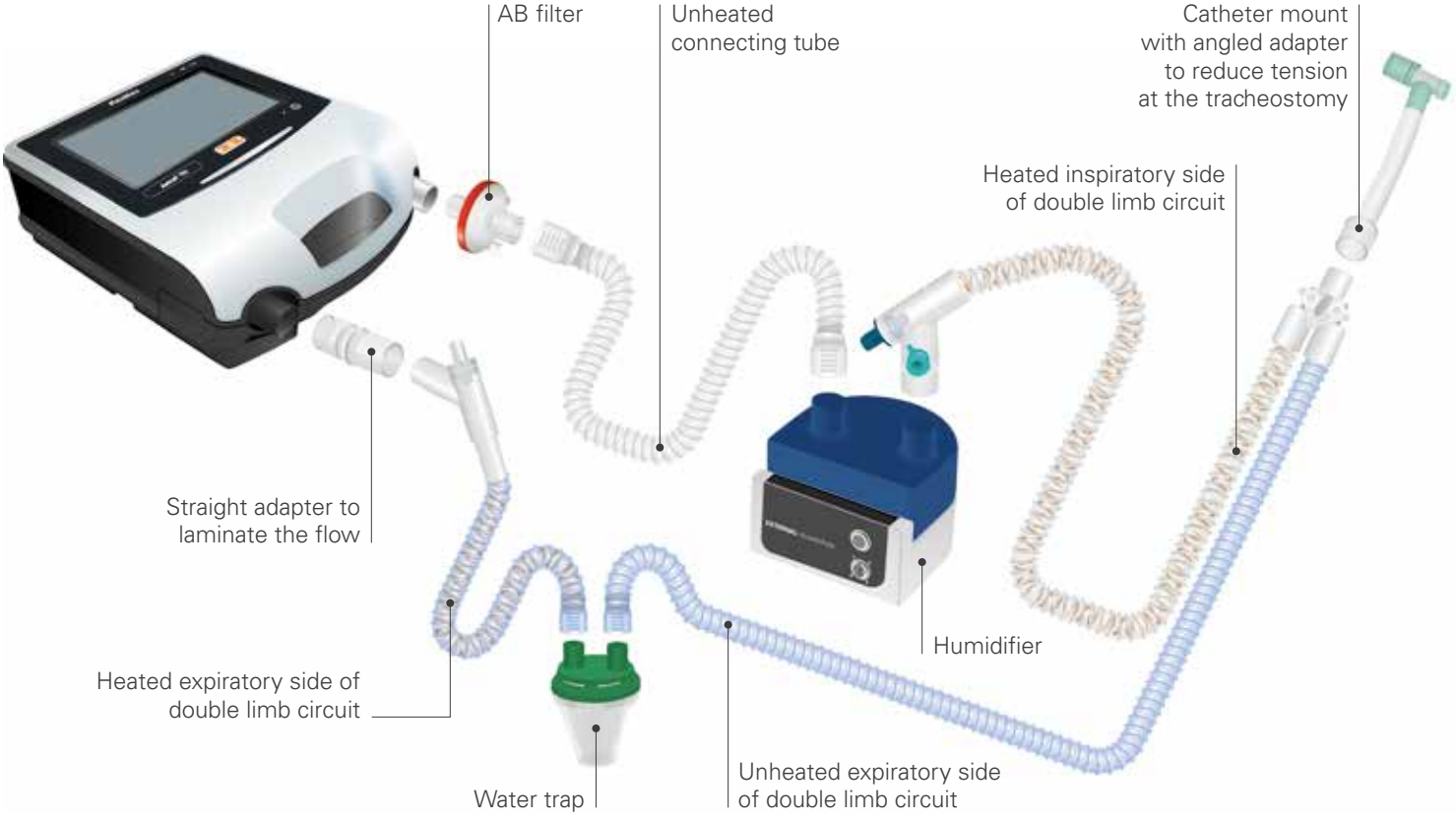


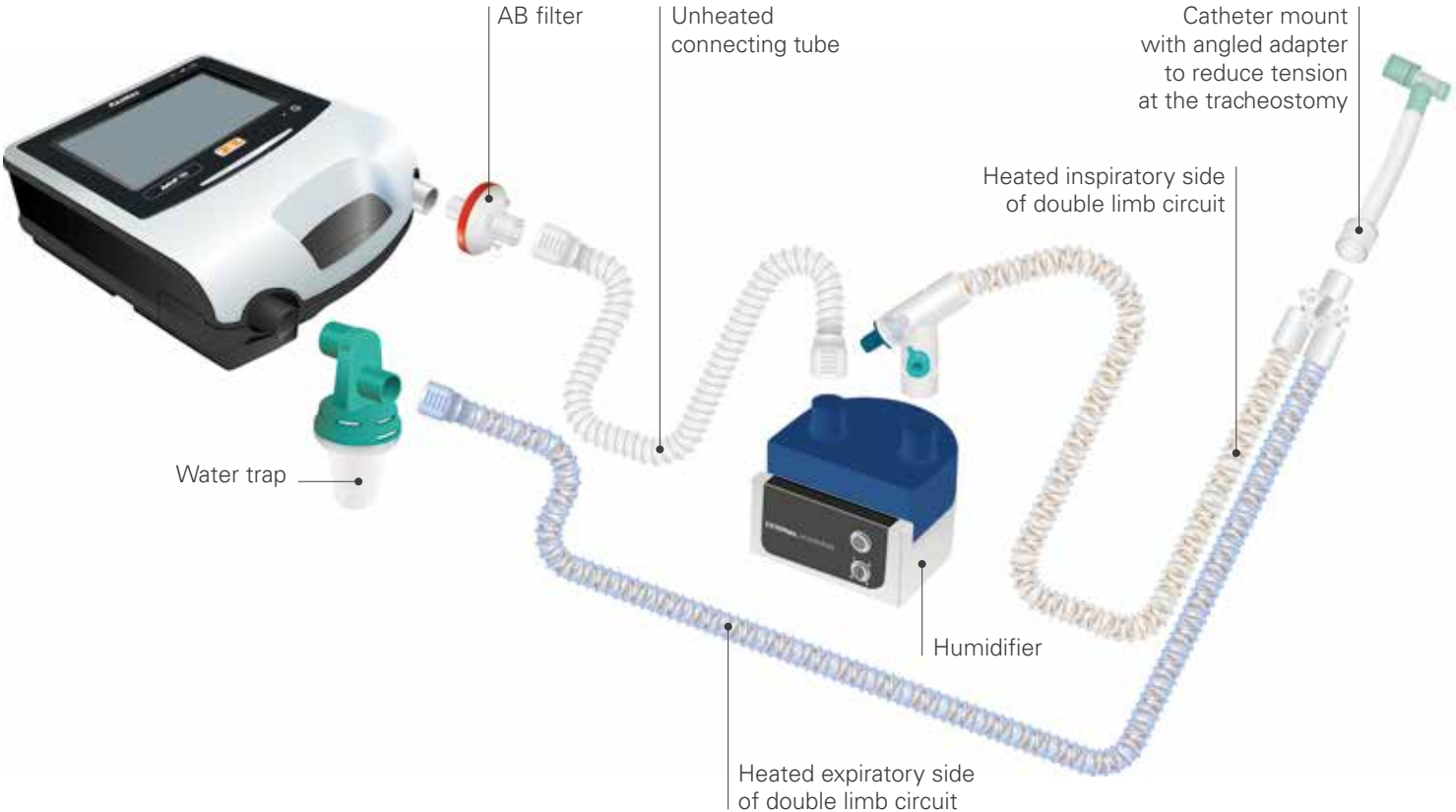
Straight or angled adapter in front of expiratory port



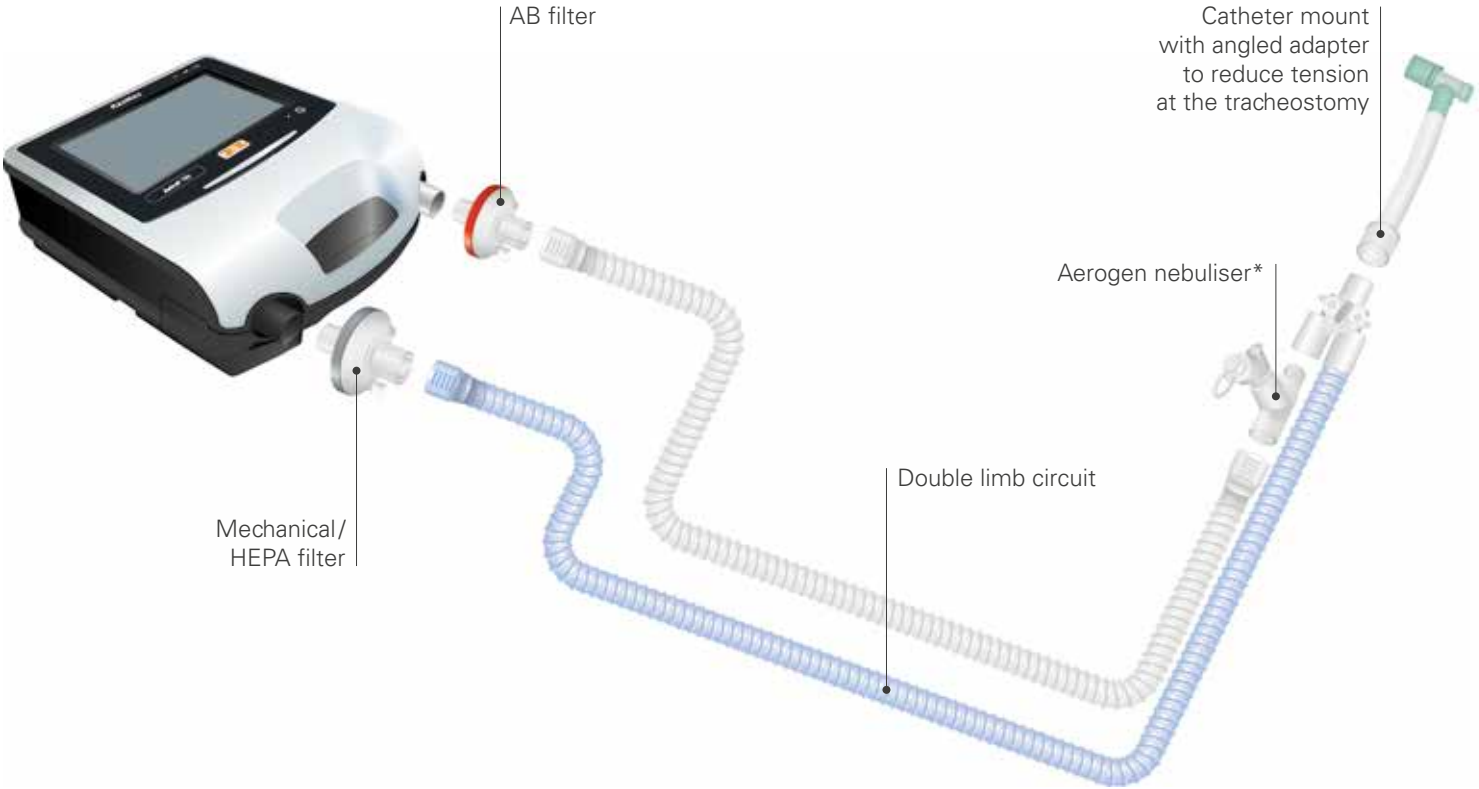

AB filter
Unheated connecting tube
Catheter mount with angled adapter to reduce tension at the tracheostomy
Heated inspiratory side of double limb circuit
Mechanical/ HEPA filter
Humidifier
An additional angled adapter helps to reduce water in the exhalation valve
Heated expiratory side of double limb circuit


AB filter
Unheated
connecting tube
Catheter mount
with angled adapter
to reduce tension
at the tracheostomy
Heated inspiratory side
of double limb circuit
Humidifier
Angled or straight adapter
to laminate the flow
Heated expiratory side of double
limb circuit with membrane


AB filter
Unheated
connecting tube
Catheter mount
with angled adapter
to reduce tension
at the tracheostomy
Heated inspiratory side
of double limb circuit
Straight adapter to
laminate the flow
Humidifier
Heated expiratory side of
double limb circuit
Water trap
Unheated expiratory side
of double limb circuit


AB filter
Unheated
connecting tube
Catheter mount
with angled adapter
to reduce tension
at the tracheostomy
Heated inspiratory side
of double limb circuit
Water trap
Humidifier
Heated expiratory side
of double limb circuit

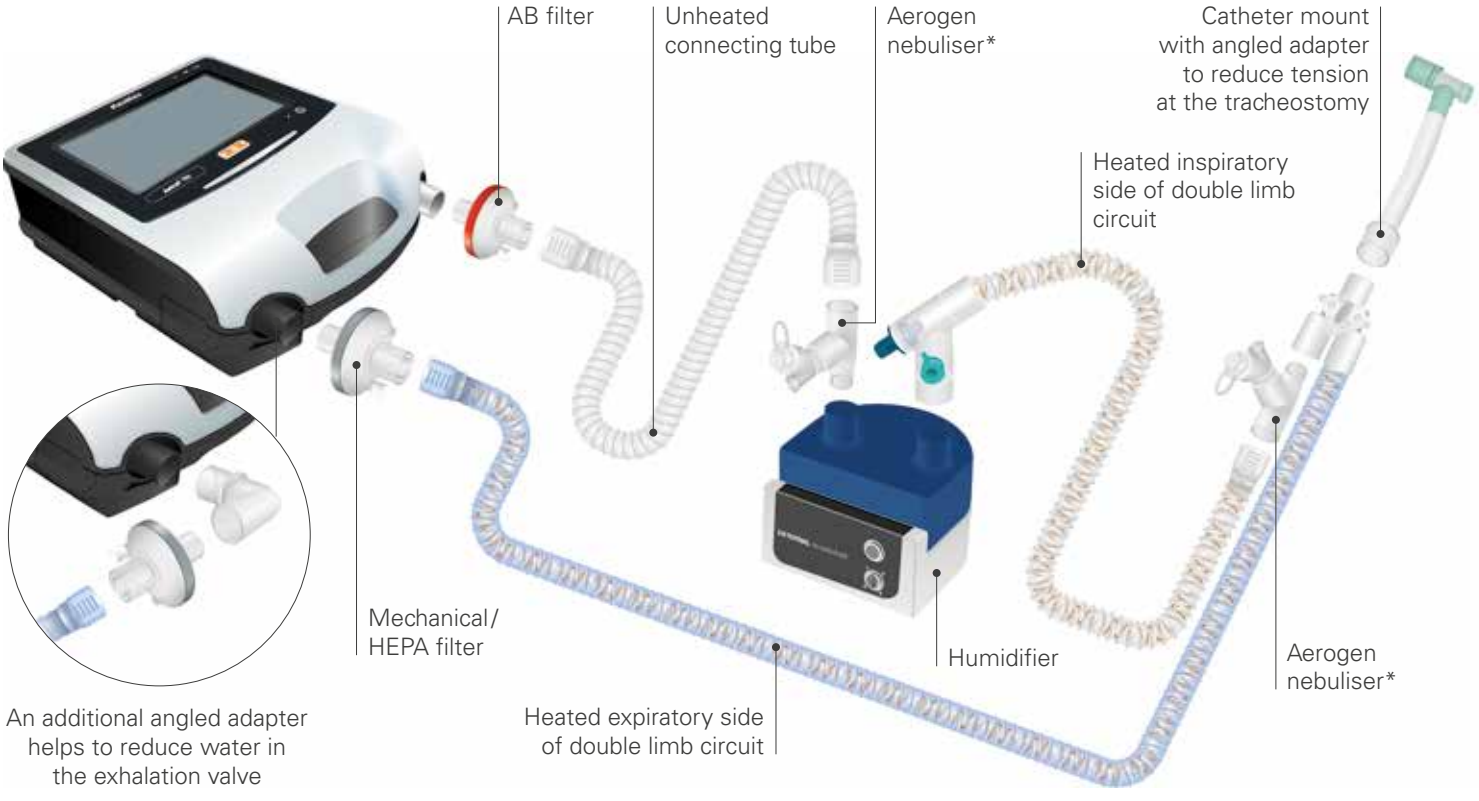
HEPA filter in front of expiratory port



* Please see Aerogen user guide for details.

Note: Adding a HEPA filter at the air inlet helps to prevent inhalation deposits in the exhalation valve during nebulisation.

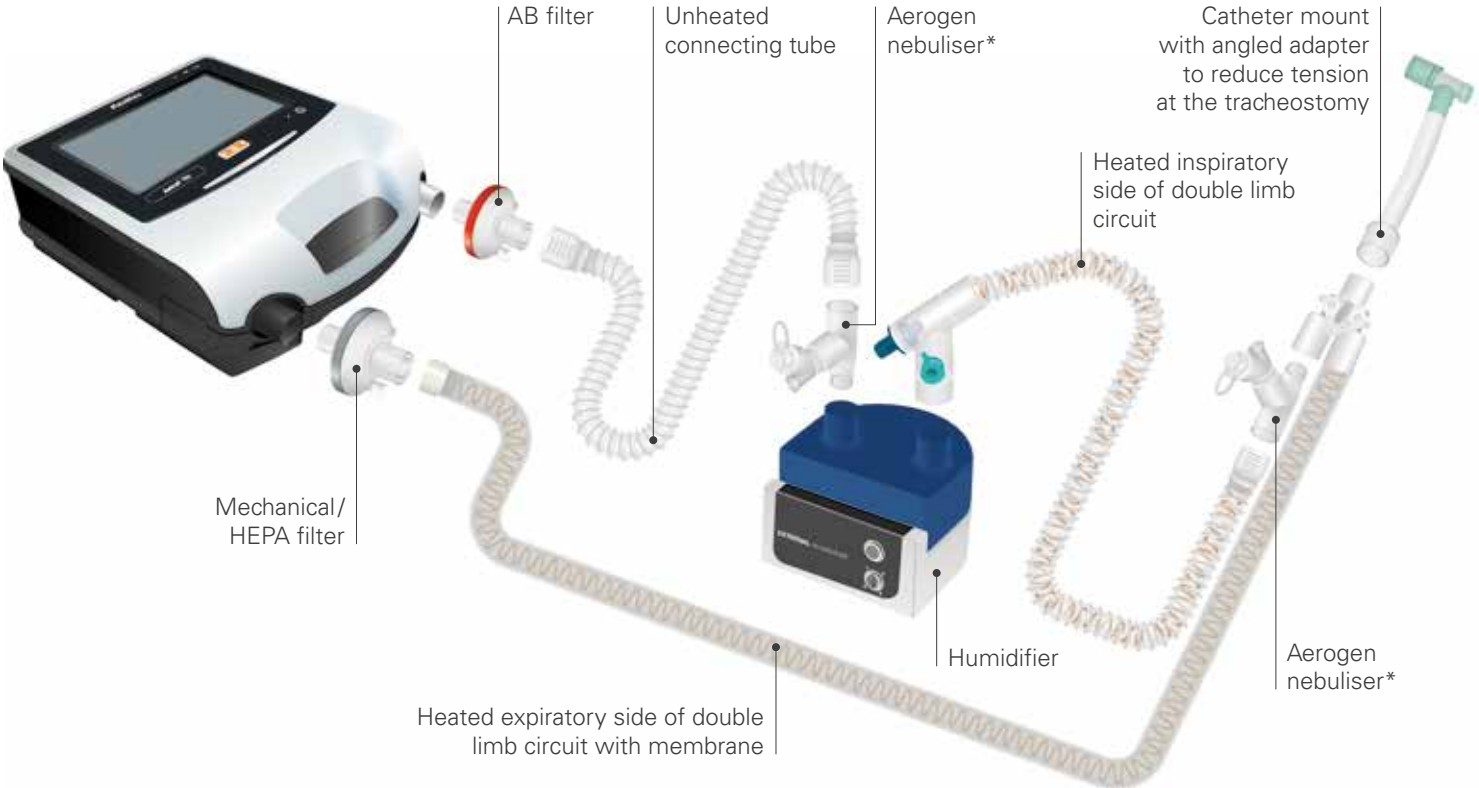
HEPA filter in front of adapter and expiratory port



* Please see Aerogen user guide for details.

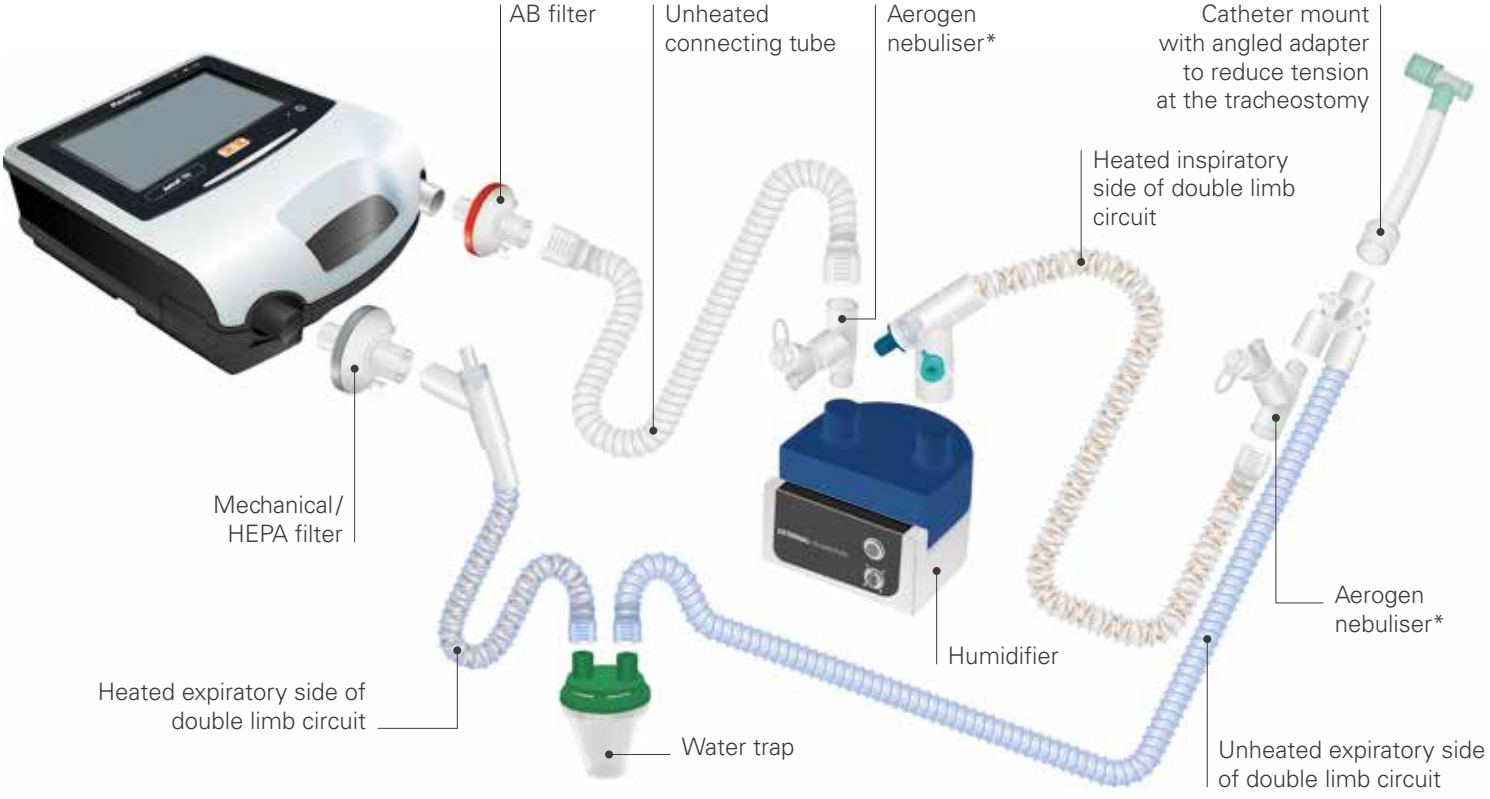
Note: Adding a HEPA filter at the air inlet helps to prevent inhalation deposits in the exhalation valve during nebulisation.

Membrane circuit and HEPA filter in front of expiratory port



* Please see Aerogen user guide for details.

Note: Adding a HEPA filter at the air inlet helps to prevent inhalation deposits in the exhalation valve during nebulisation.



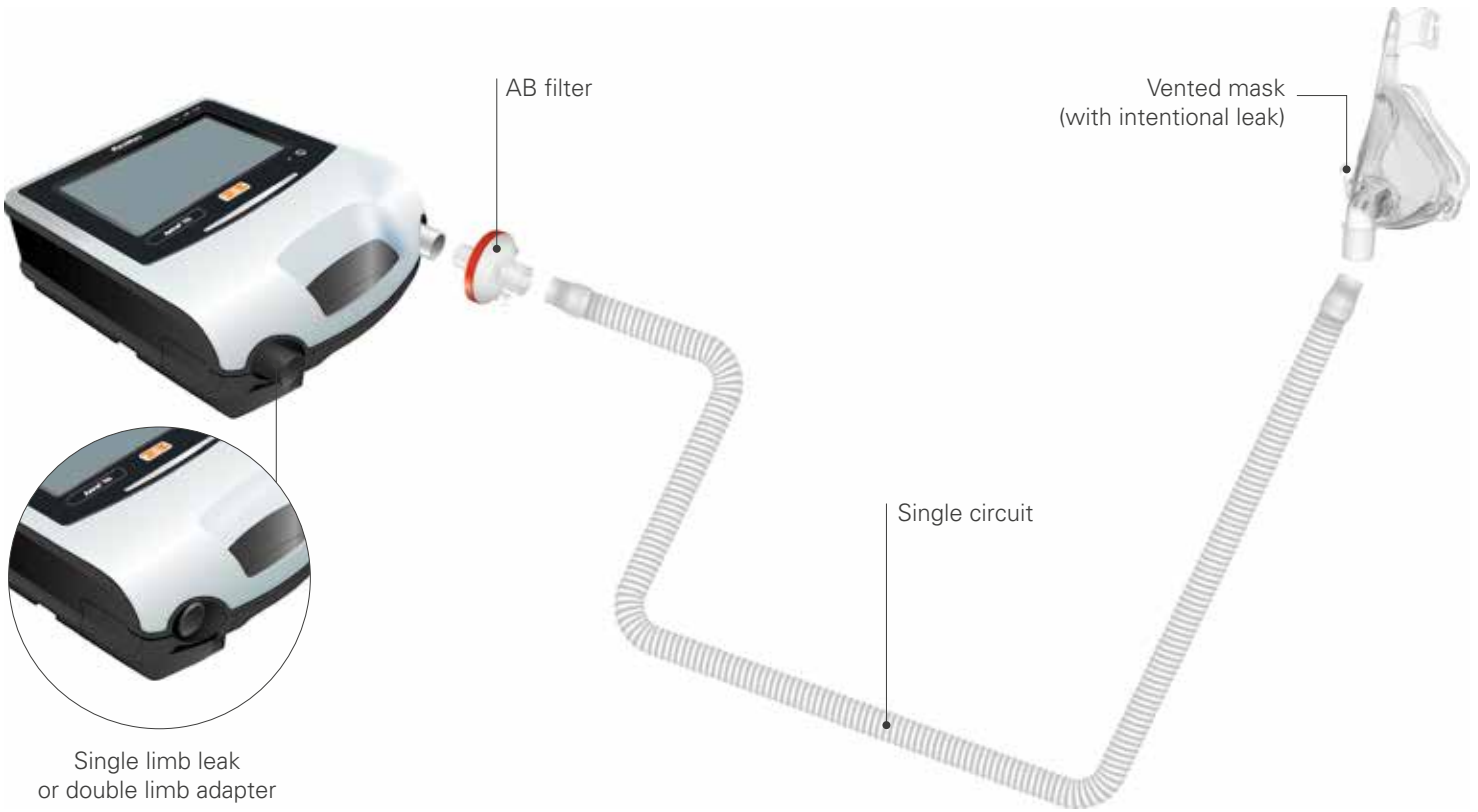
* Please see Aerogen user guide for details.

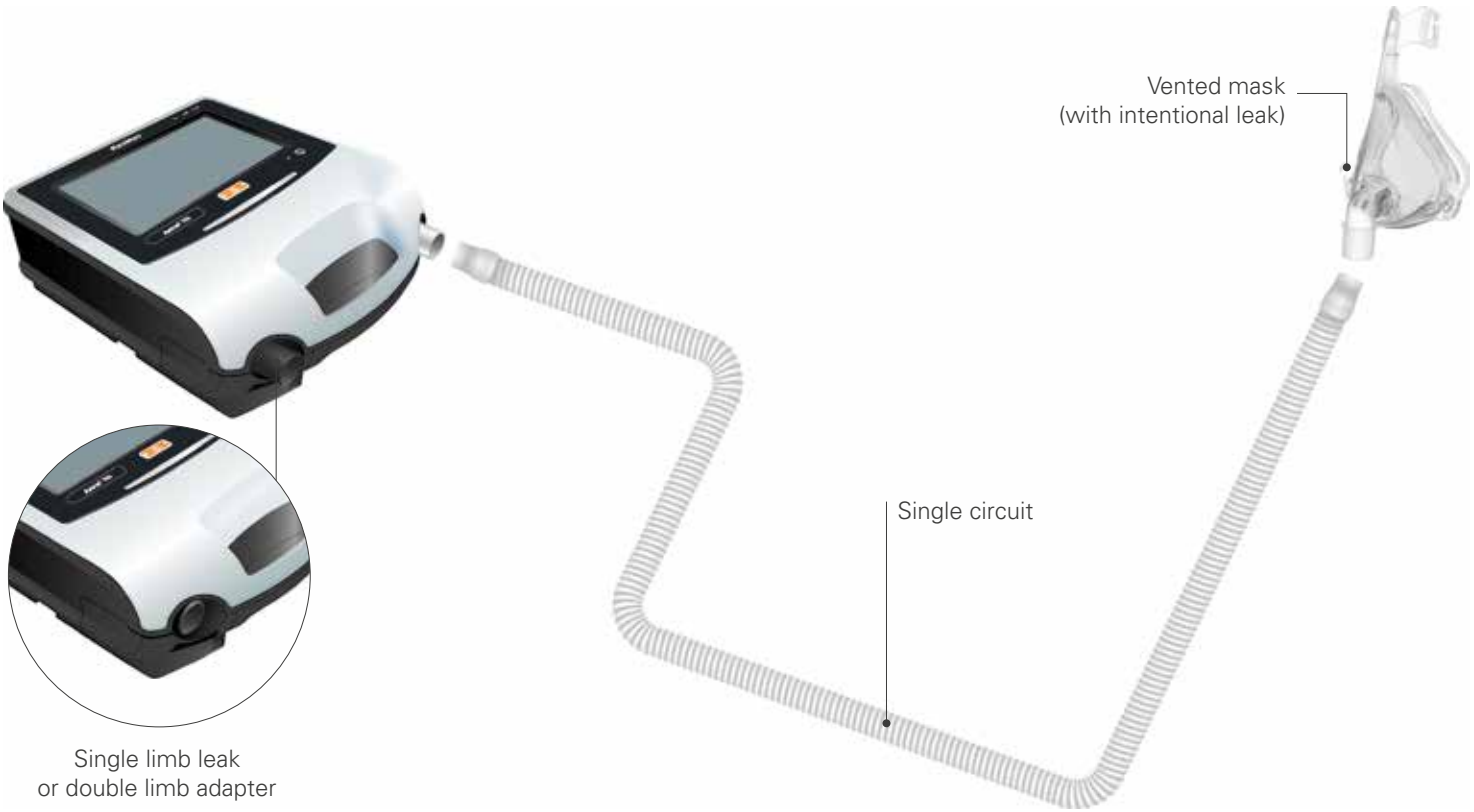
Note: Adding a HEPA filter at the air inlet helps to prevent inhalation deposits in the exhalation valve during nebulisation.

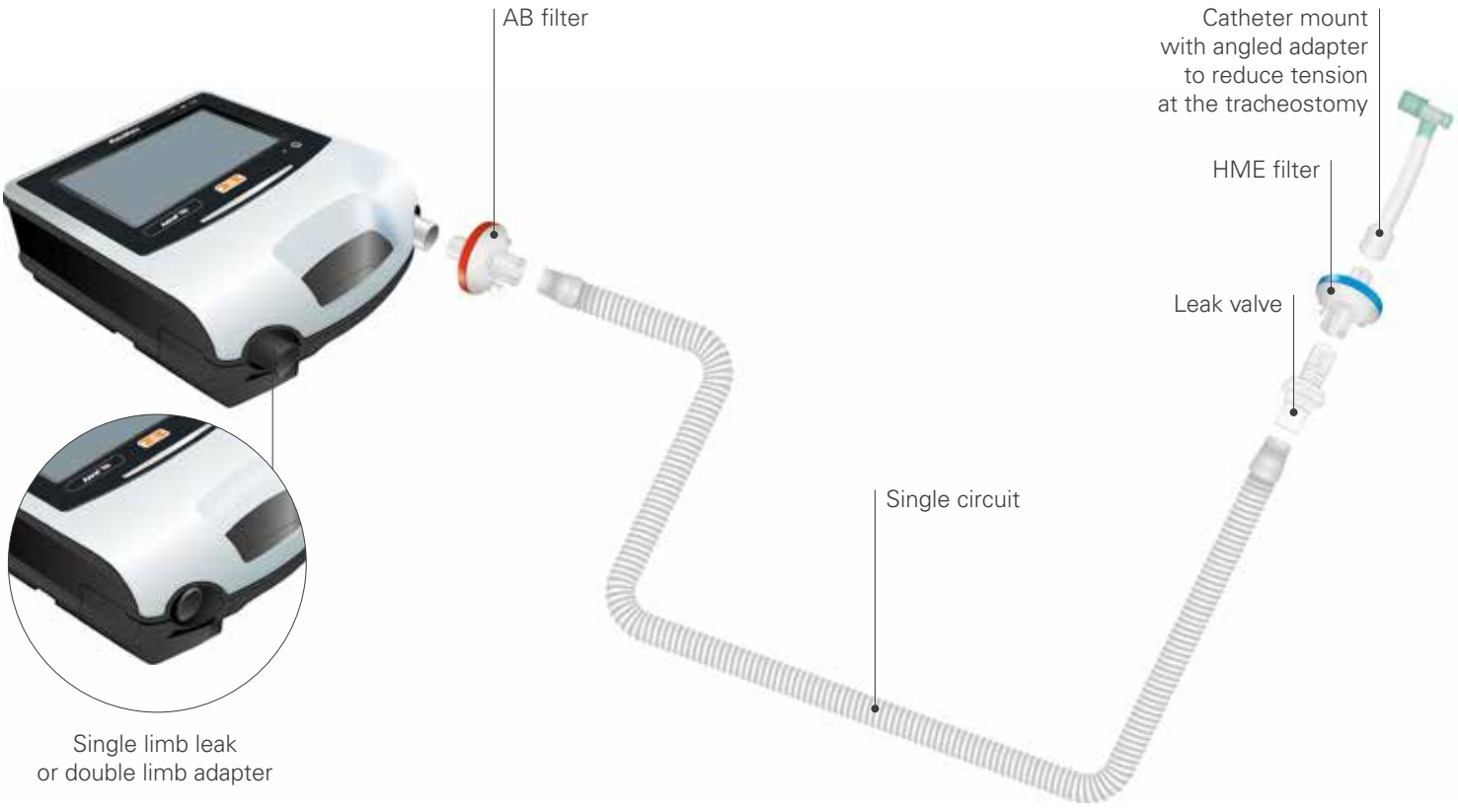
Leak circuit

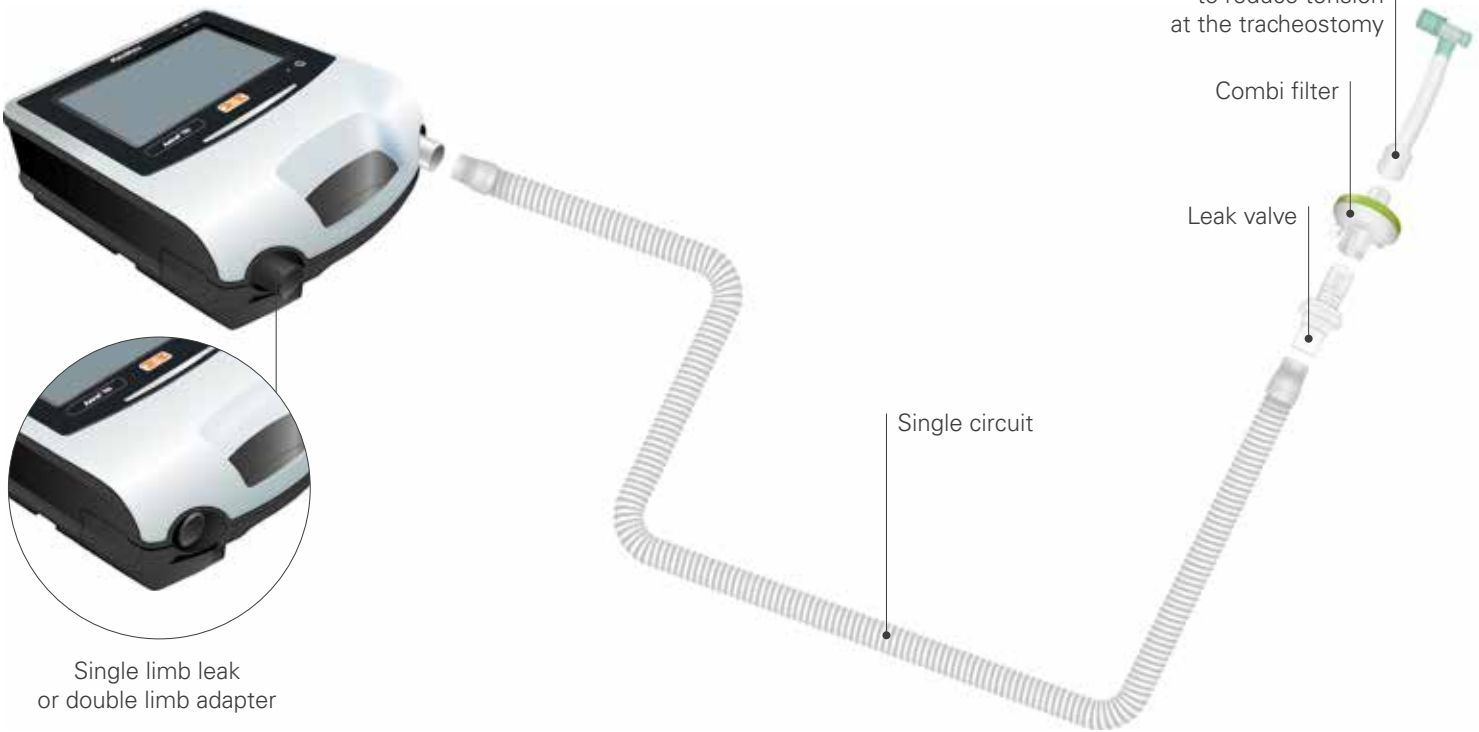


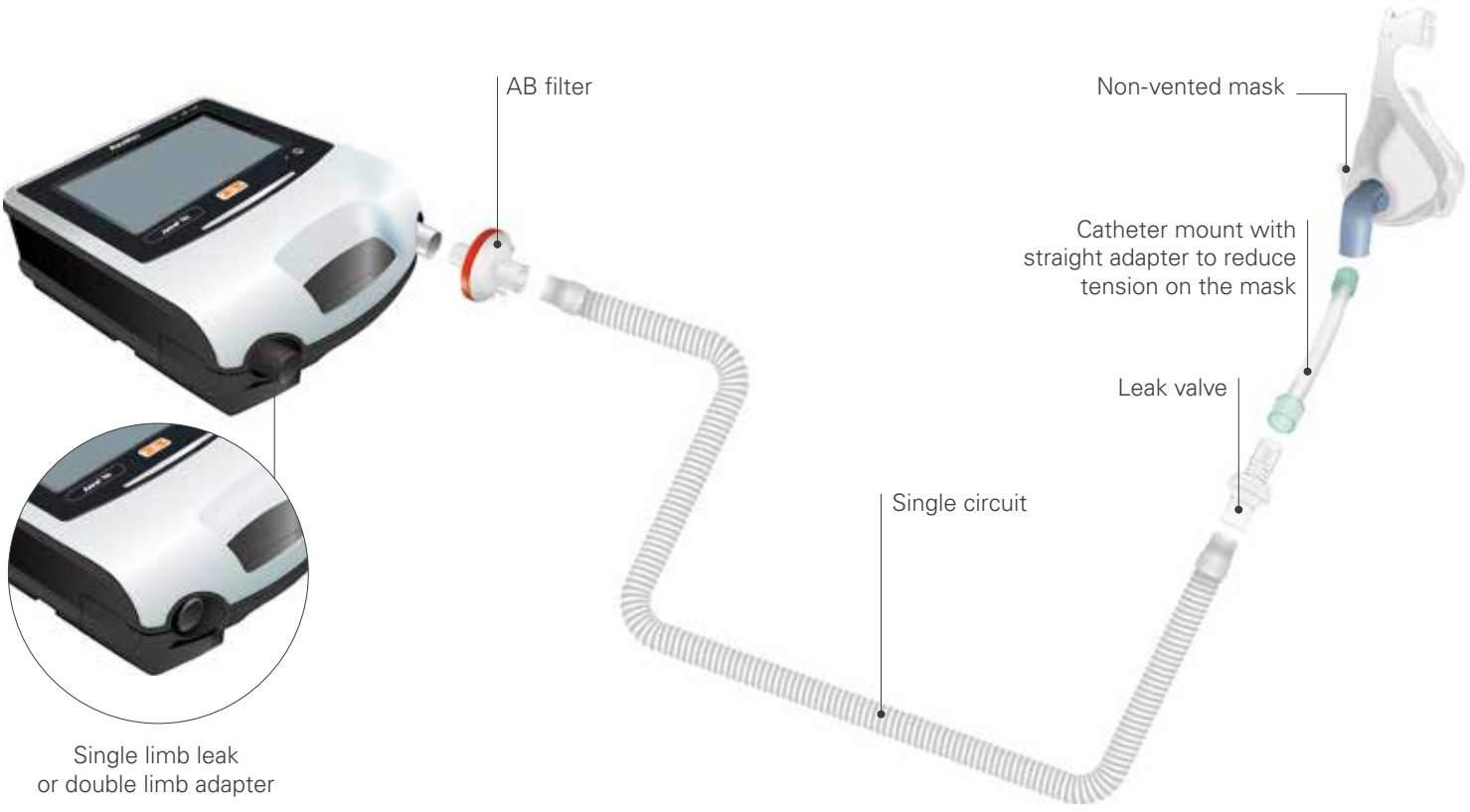
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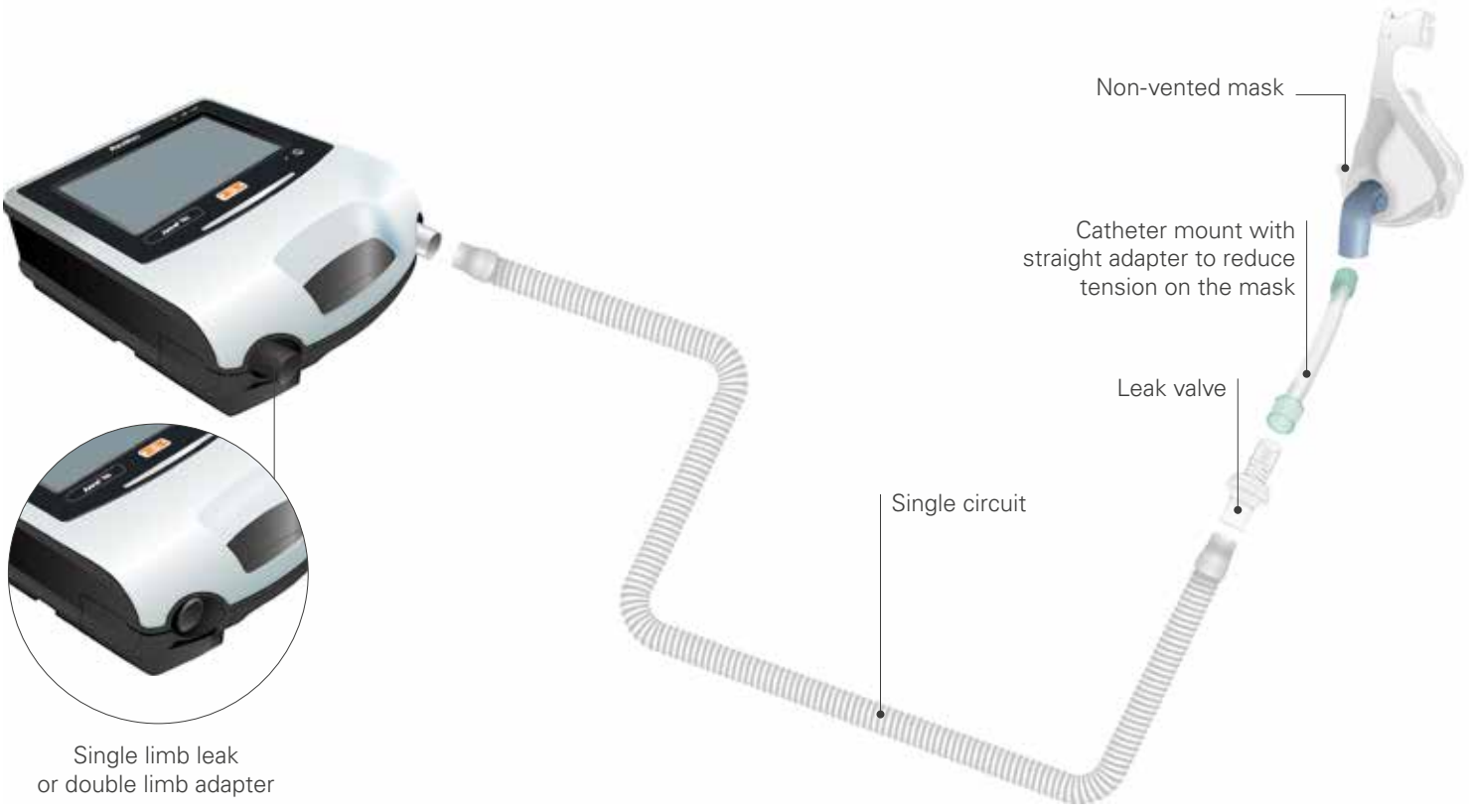

AB filter
Vented mask
(with intentional leak)
Single circuit
Single limb leak
or double limb adapter

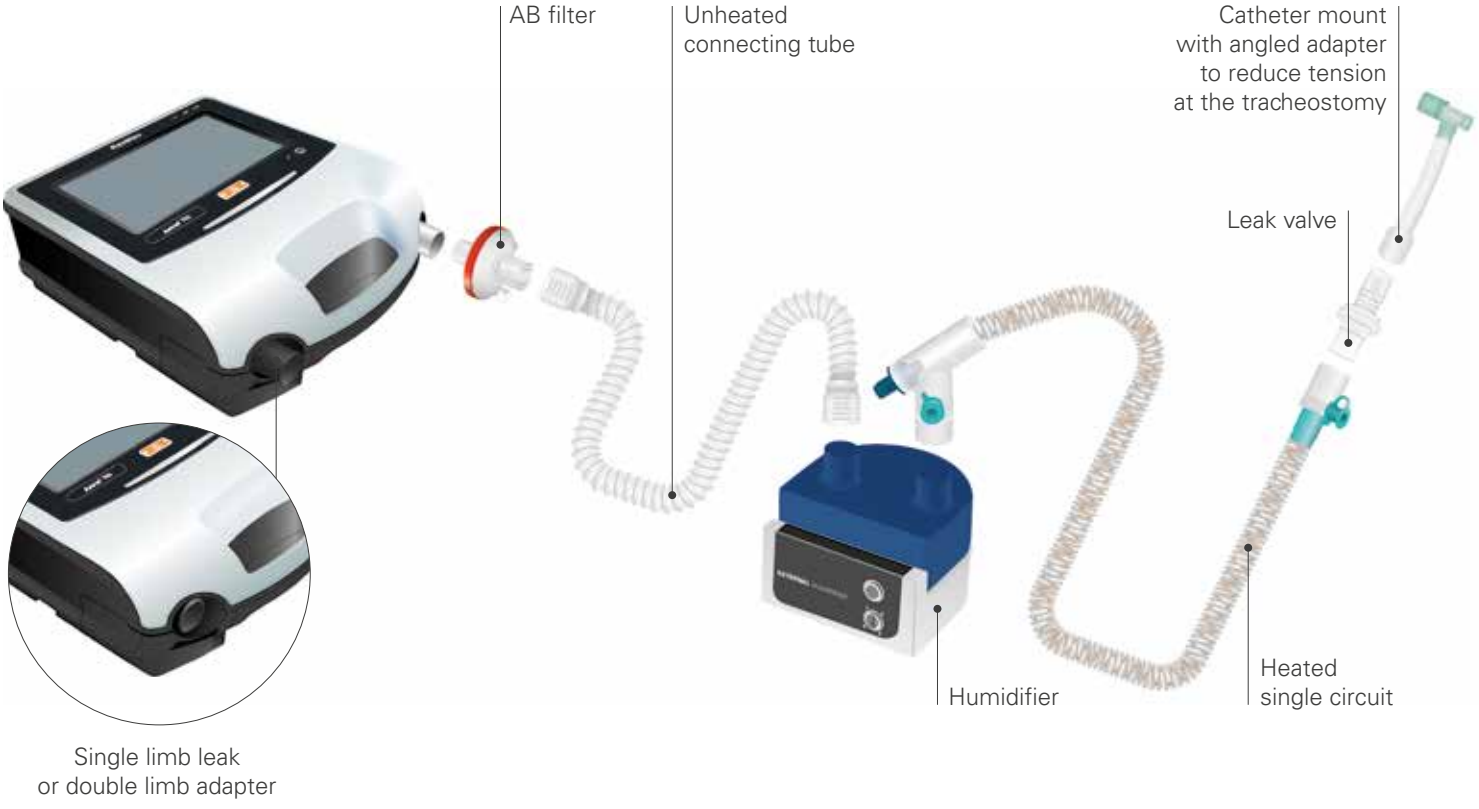

Vented mask
(with intentional leak)
Single circuit
Single limb leak
or double limb adapter

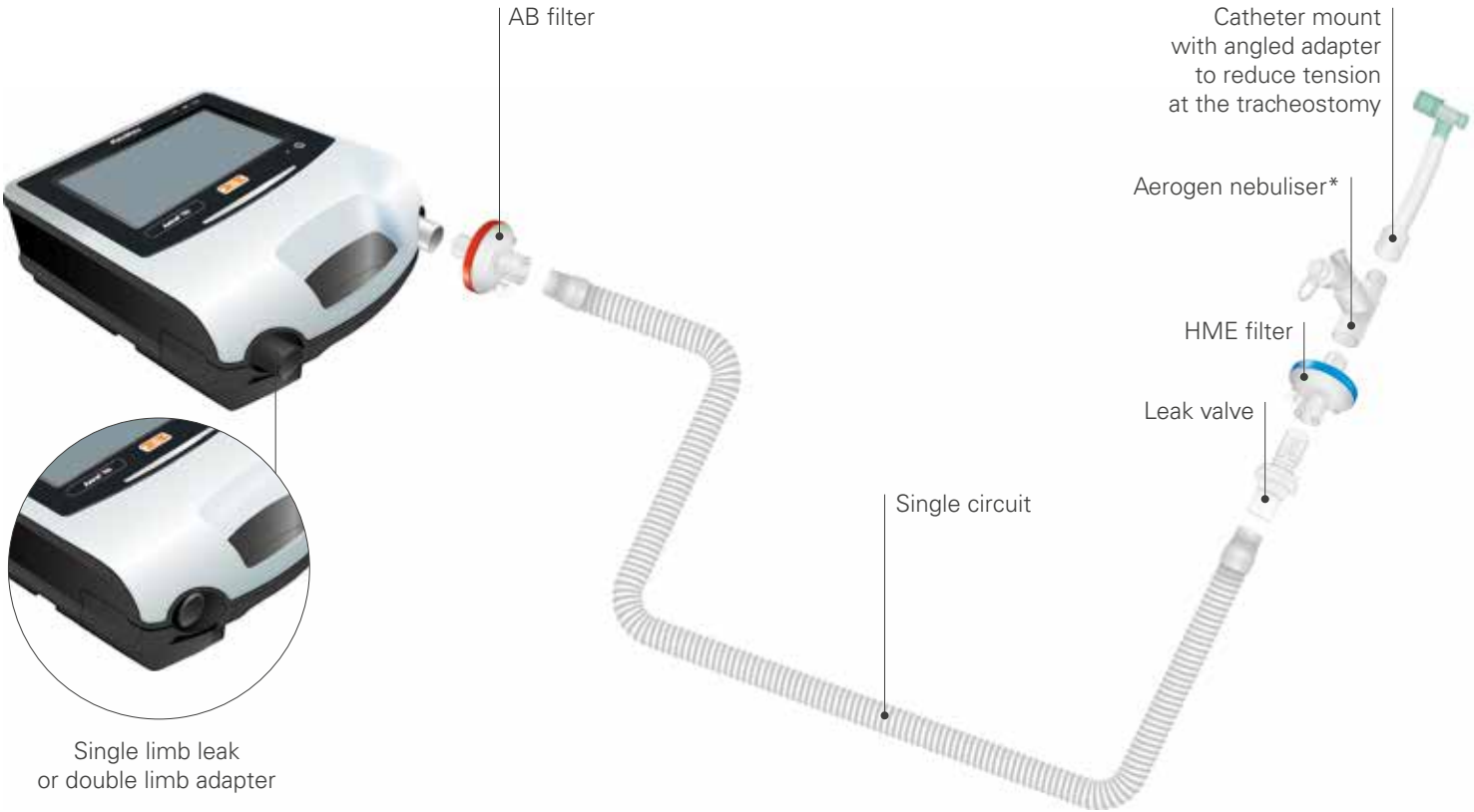

AB filter
Catheter mount
with angled adapter
to reduce tension
at the tracheostomy
HME filter
Leak valve
Single circuit
Single limb leak
or double limb adapter


Catheter mount
with angled adapter
to reduce tension
at the tracheostomy
Combi filter
Leak valve
Single circuit
Single limb leak
or double limb adapter

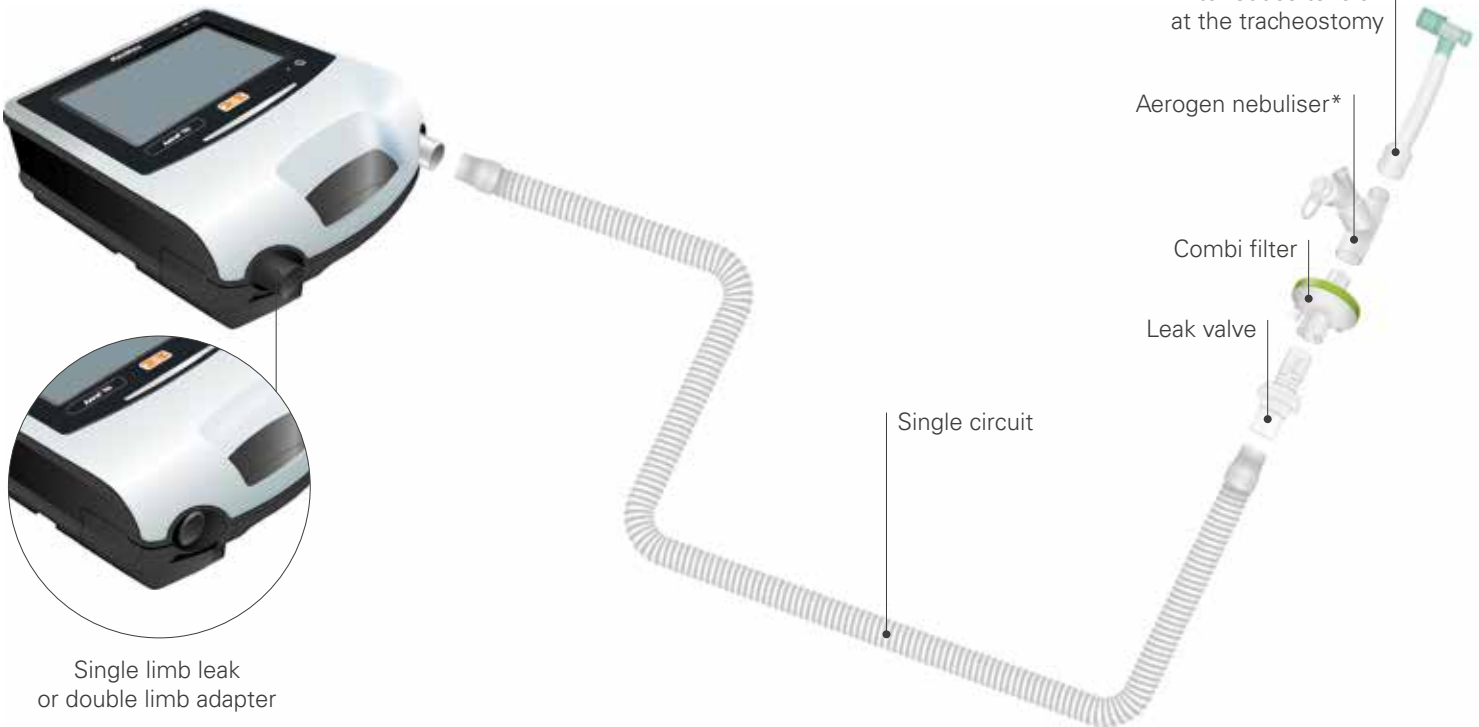

AB filter
Non-vented mask
Catheter mount with straight adapter to reduce tension on the mask
Leak valve
Single circuit
Single limb leak or double limb adapter


Non-vented mask
Catheter mount with straight adapter to reduce tension on the mask
Leak valve
Single circuit
Single limb leak or double limb adapter

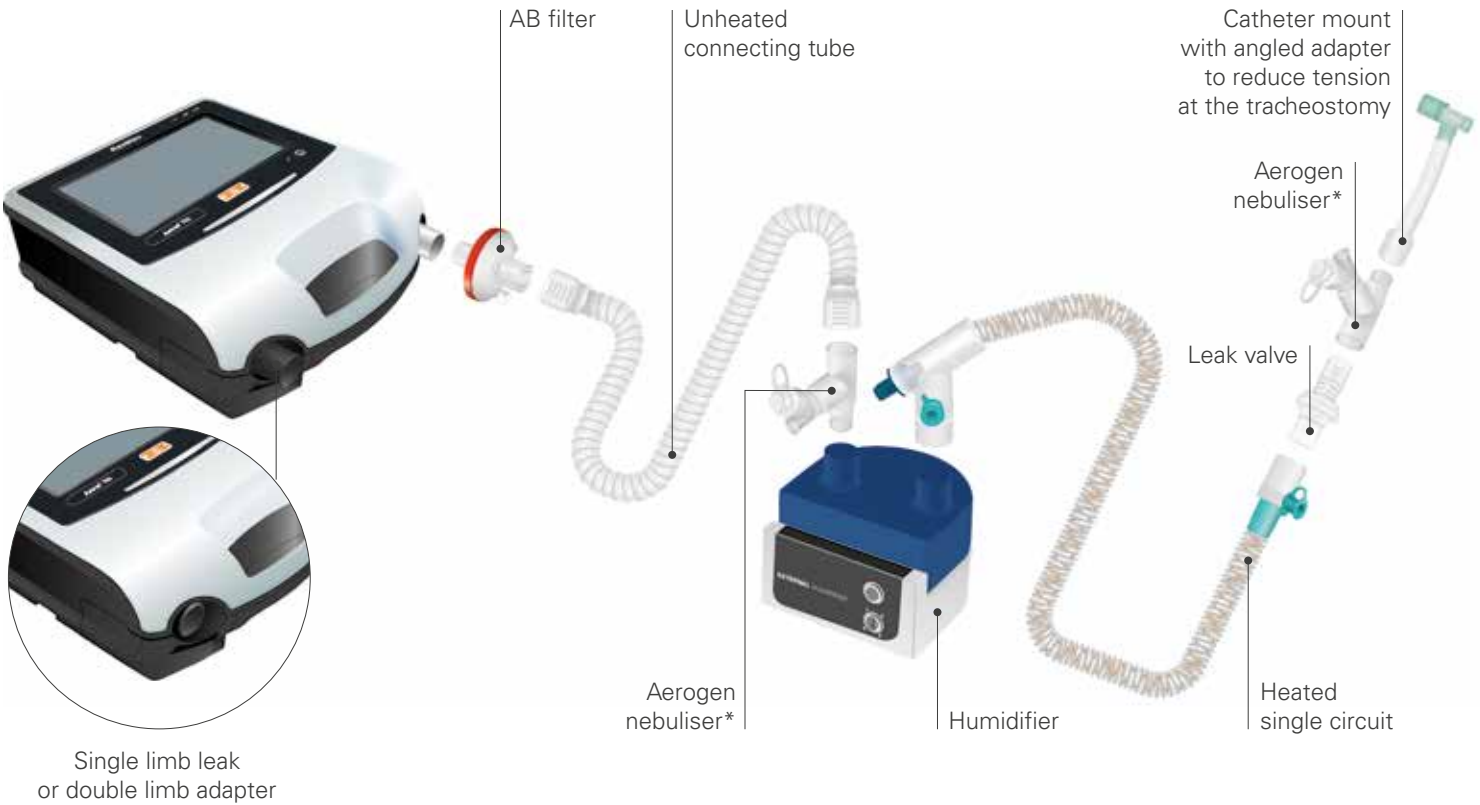

AB filter
Unheated
connecting tube
Catheter mount
with angled adapter
to reduce tension
at the tracheostomy
Leak valve
Humidifier
Heated
single circuit
Single limb leak
or double limb adapter



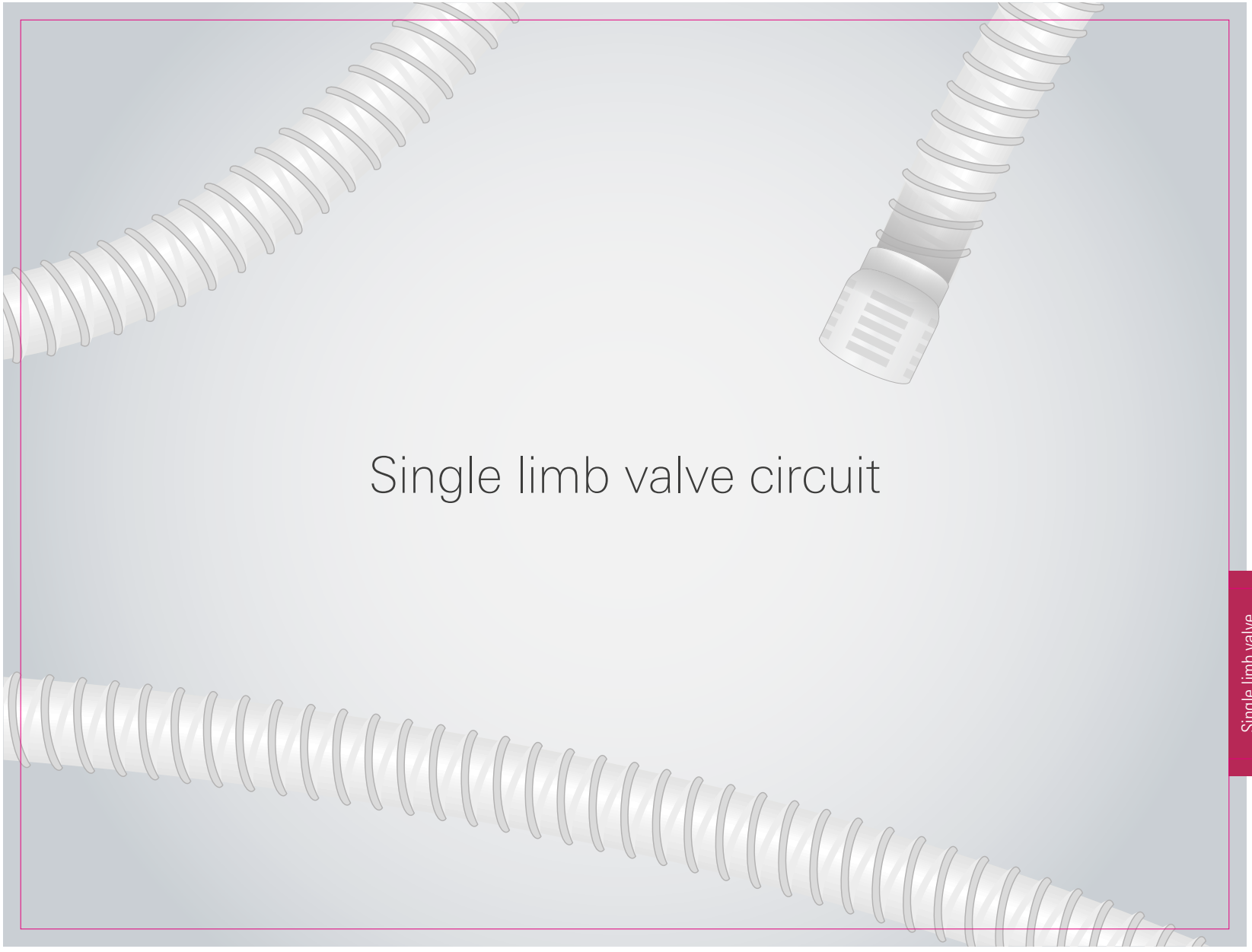
* Please see Aerogen user guide for details.



* Please see Aerogen user guide for details.



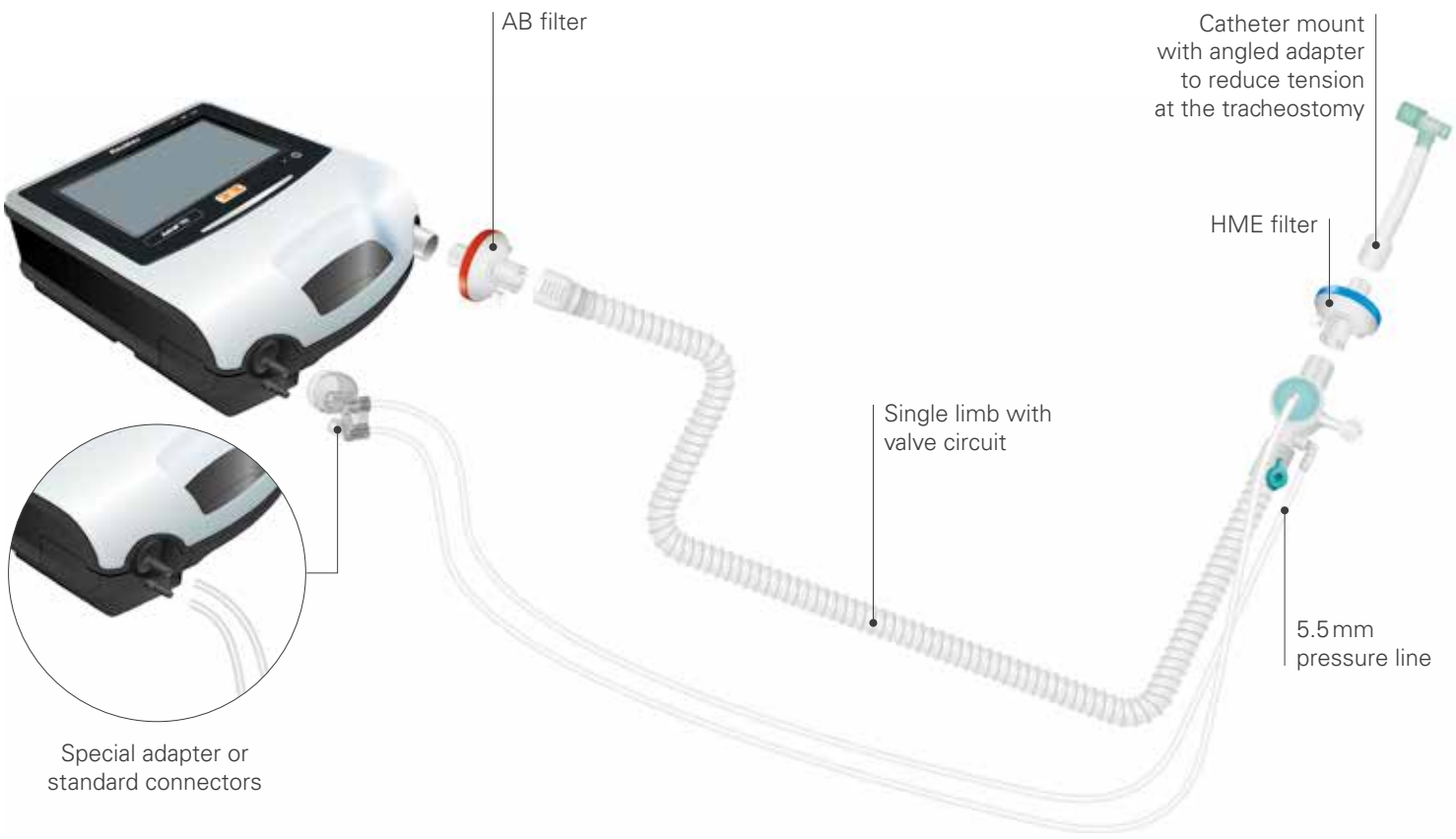
* Please see Aerogen user guide for details.

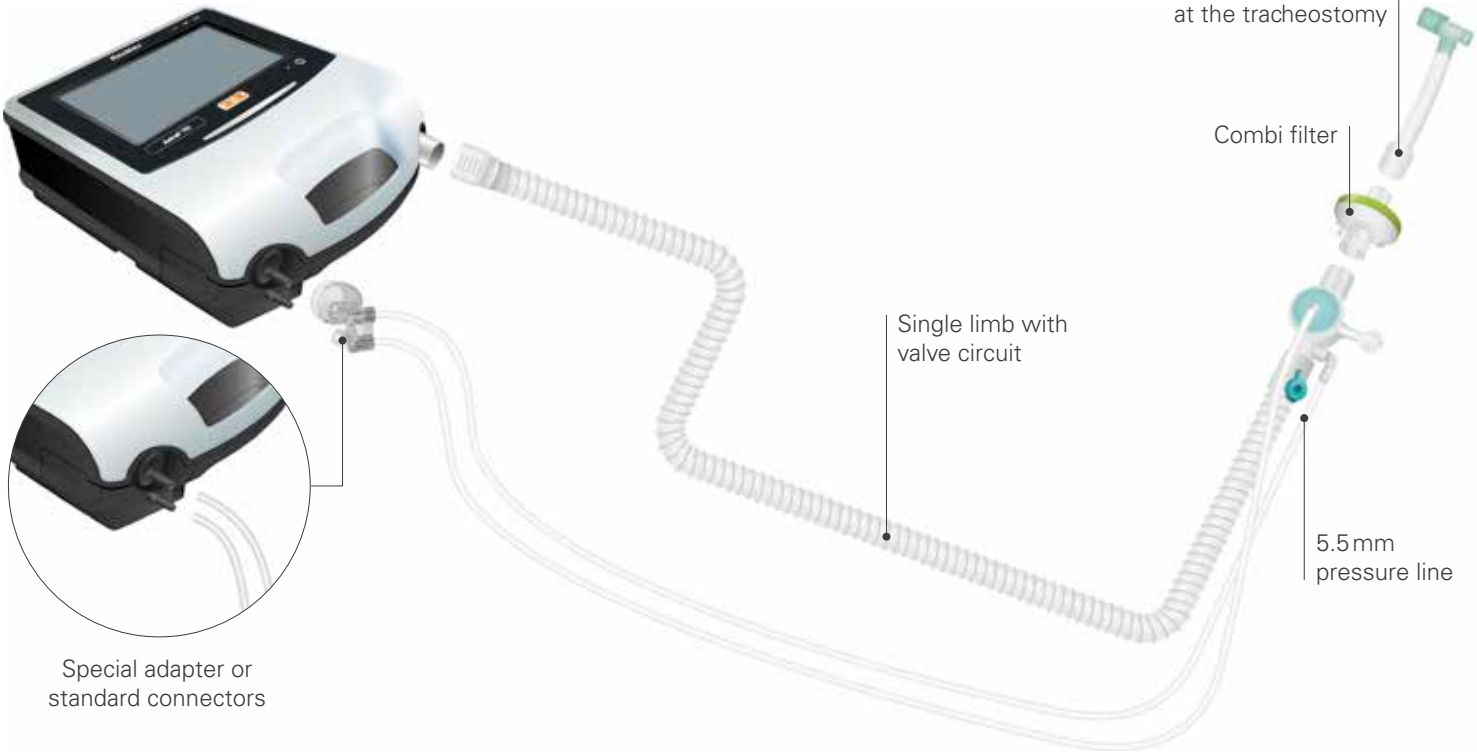


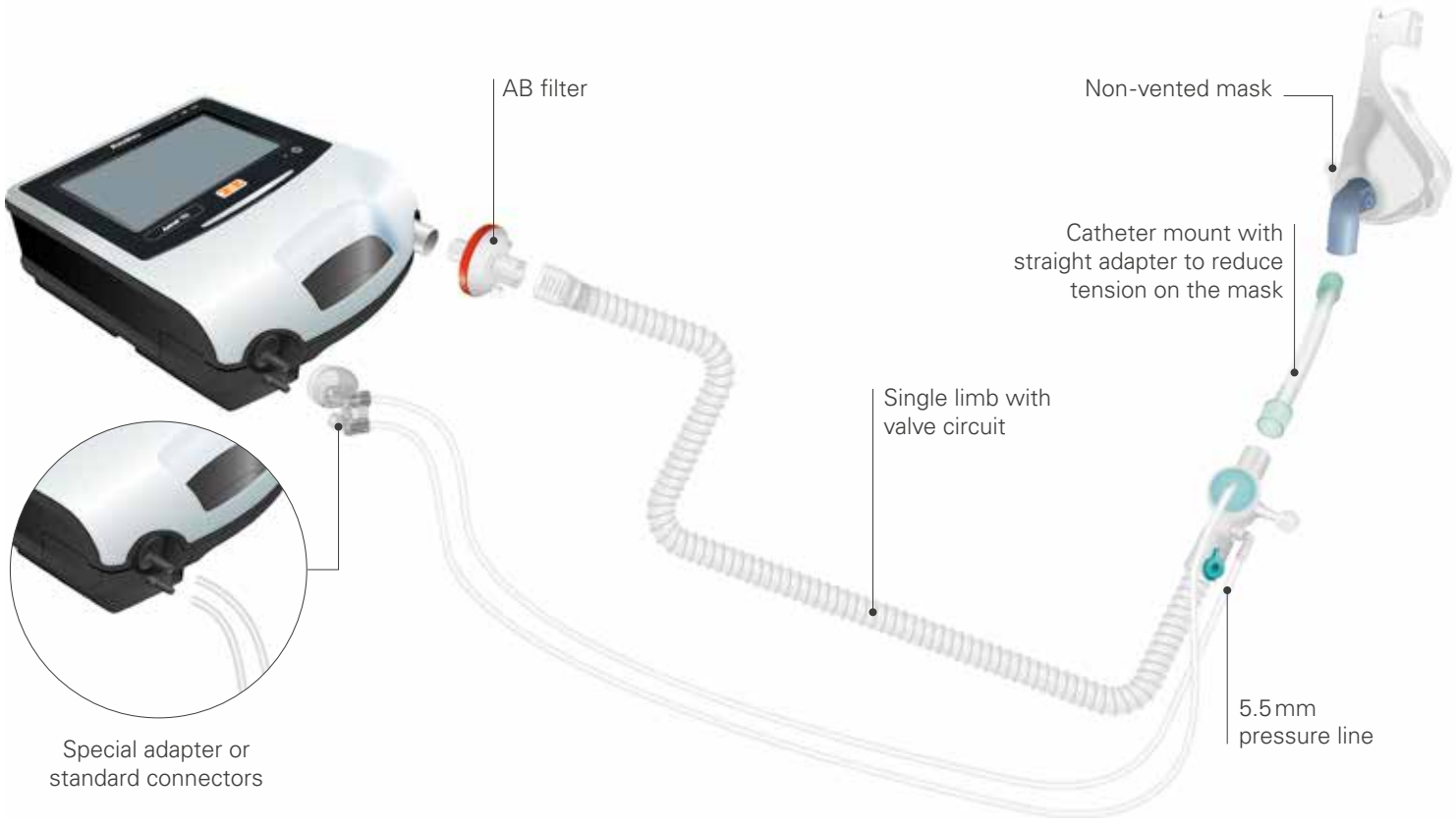
Single limb valve circuit

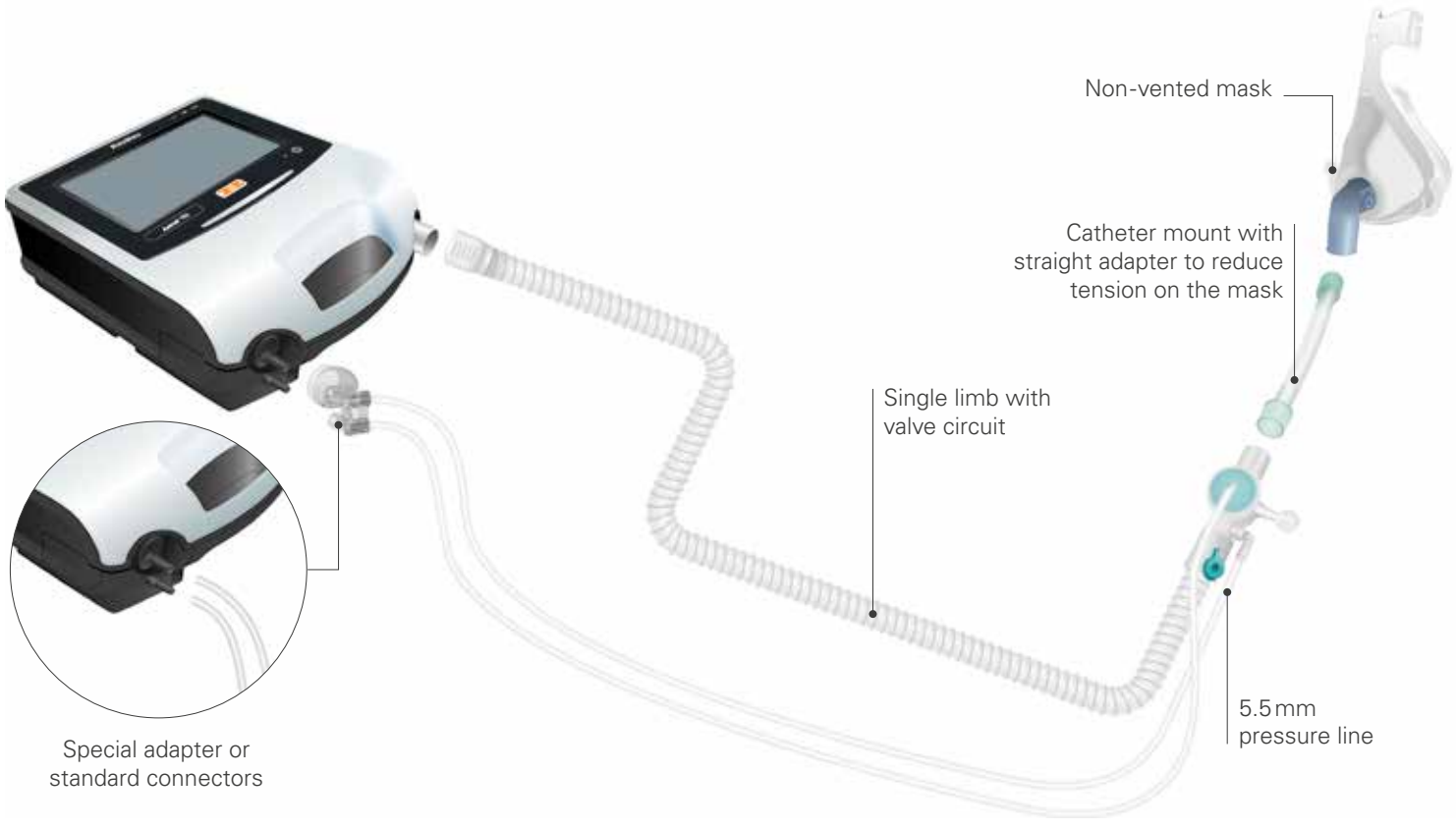
5.5 mm pressure line

1

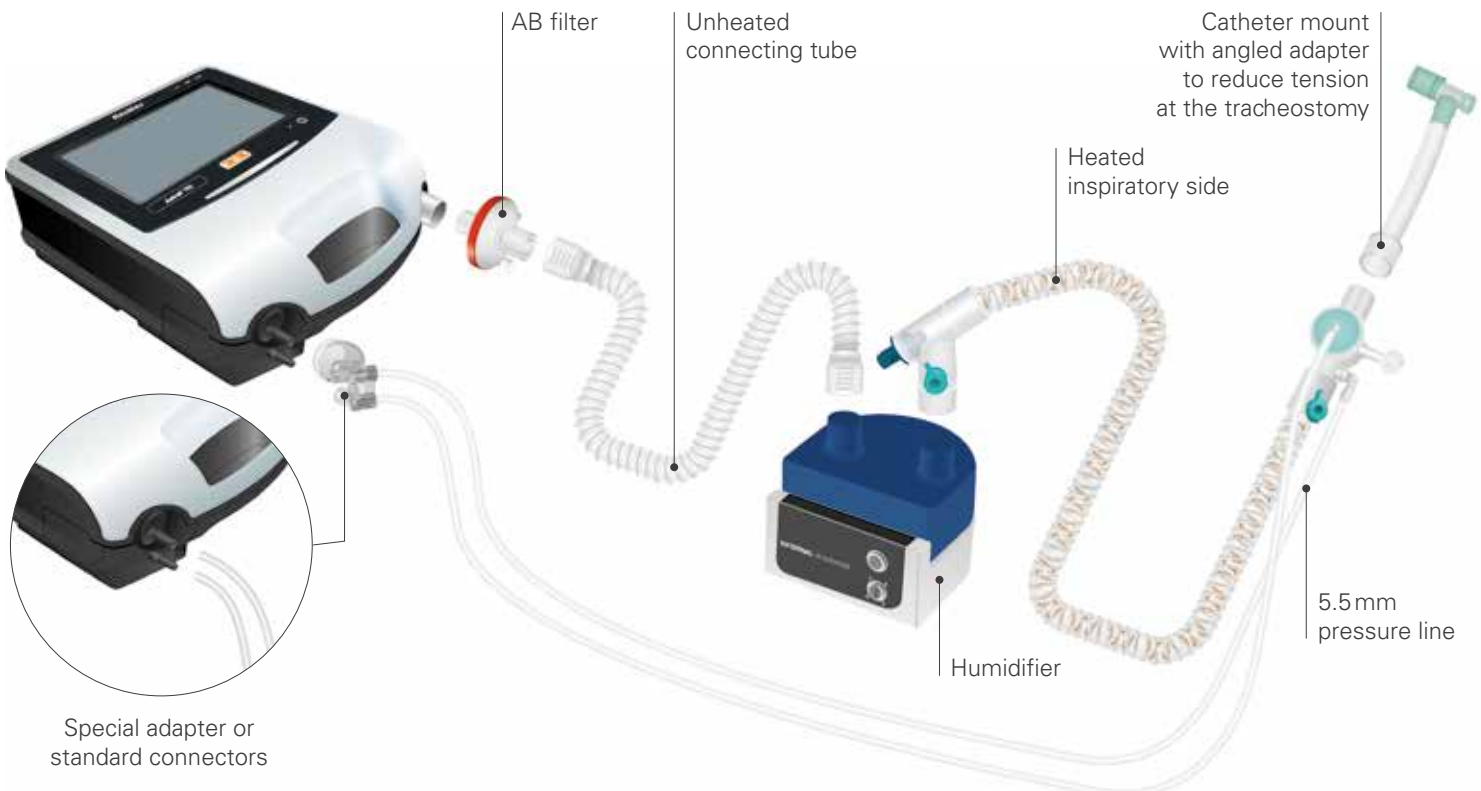



Catheter mount
with angled adapter
to reduce tension
at the tracheostomy
Combi filter
Single limb with
valve circuit
5.5 mm
pressure line
Special adapter or
standard connectors

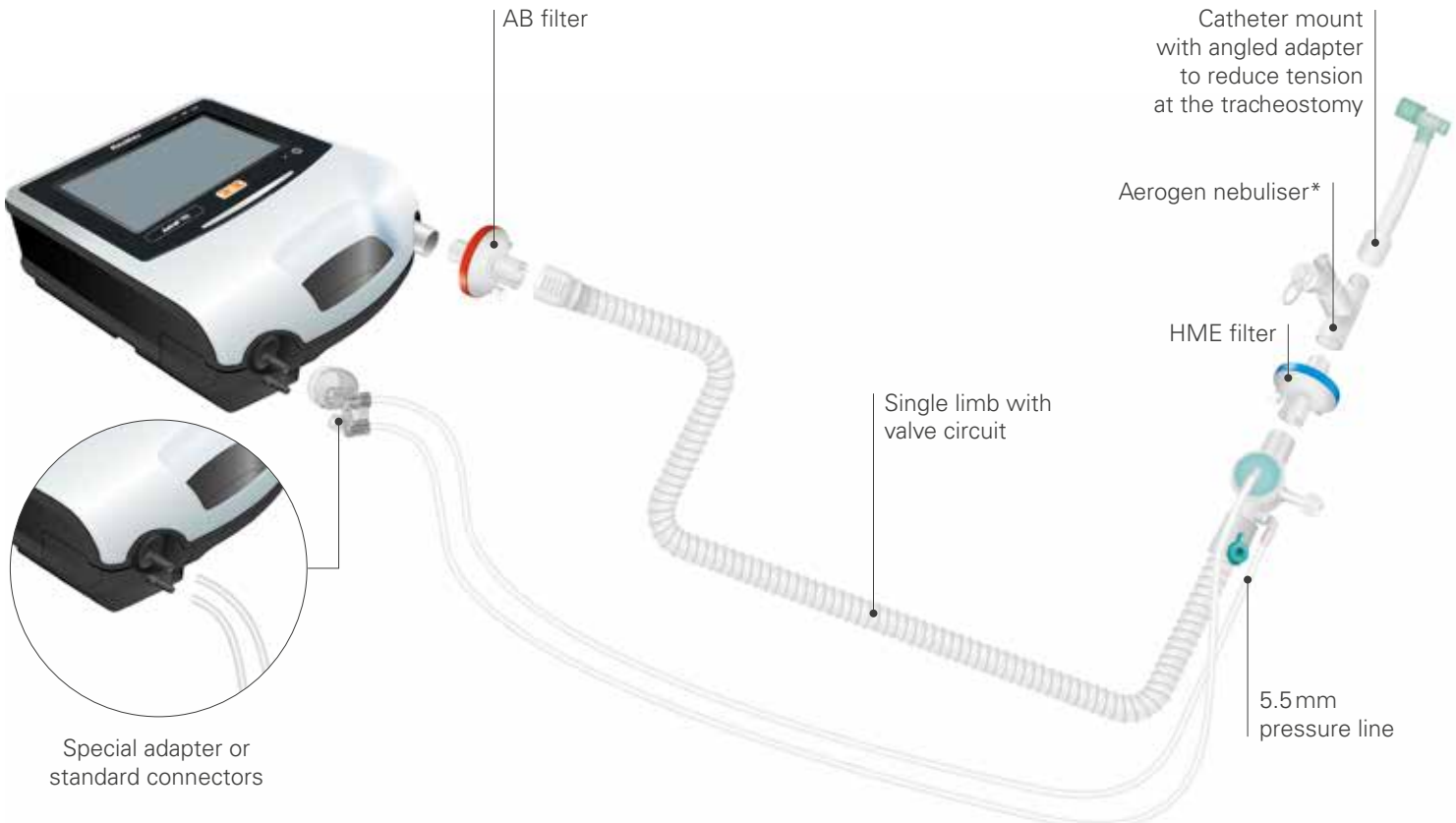

AB filter
Non-vented mask
Catheter mount with straight adapter to reduce tension on the mask
Single limb with valve circuit
5.5 mm pressure line
Special adapter or standard connectors


Non-vented mask
Catheter mount with straight adapter to reduce tension on the mask
Single limb with valve circuit
5.5 mm pressure line
Special adapter or standard connectors

5.5 mm pressure line

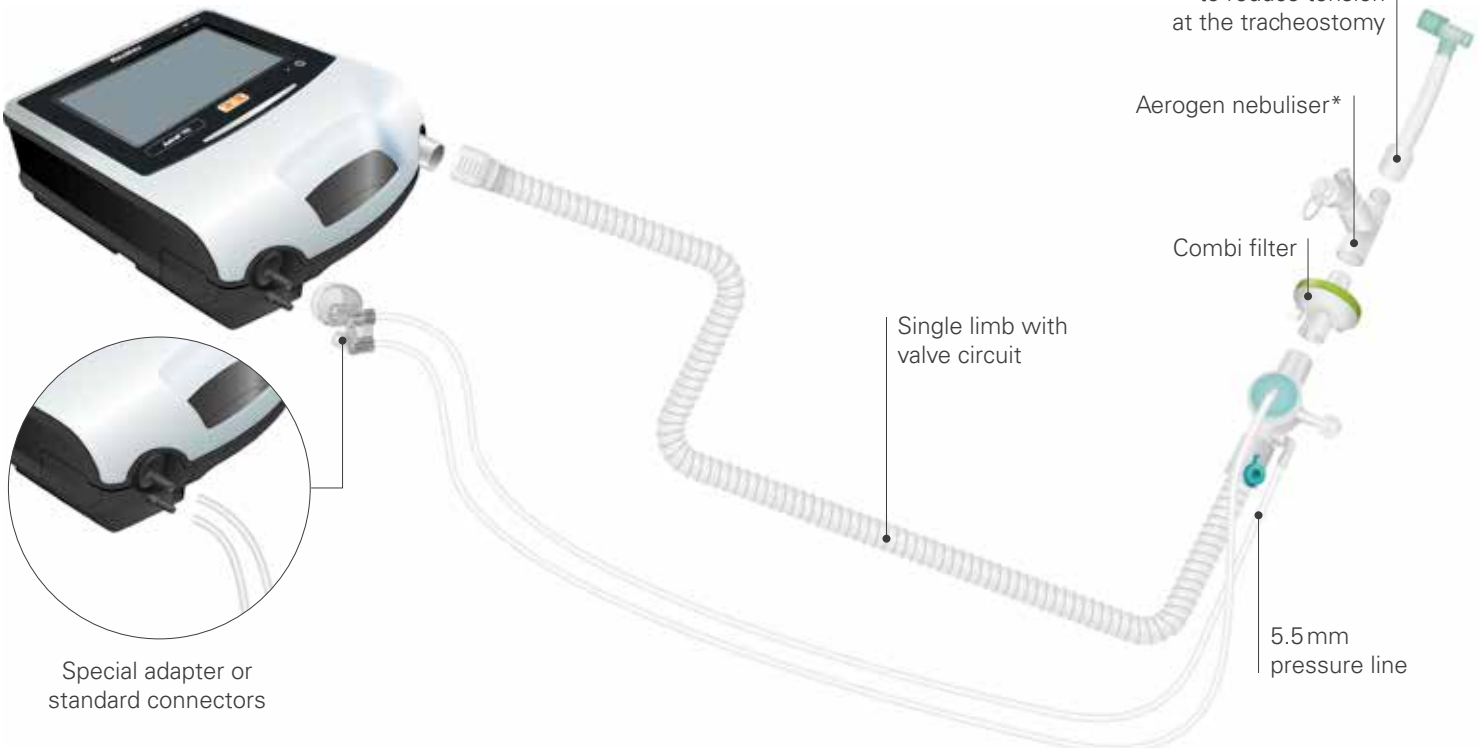


5.5 mm pressure line



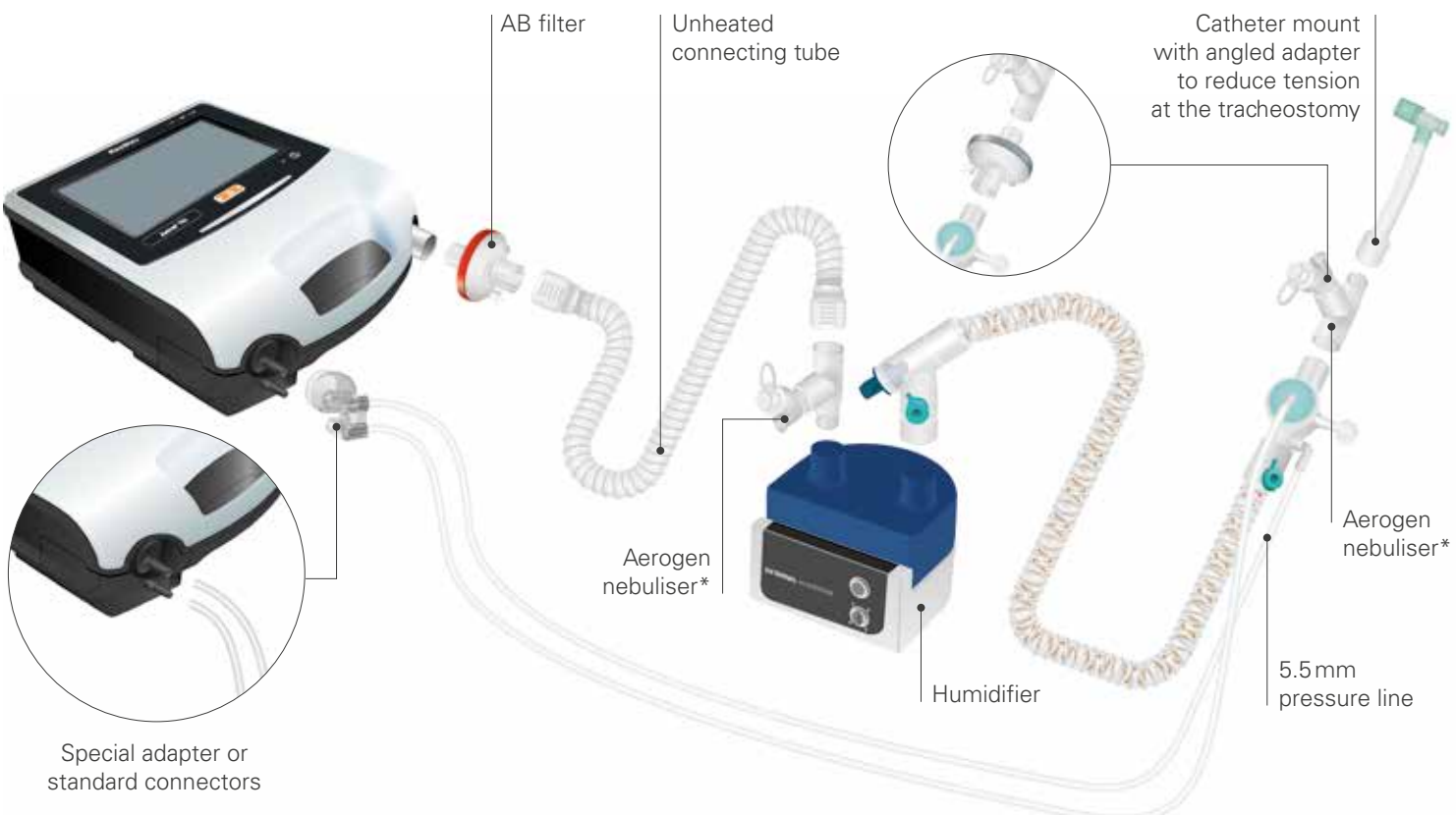
* Please see Aerogen user guide for details.

5.5 mm pressure line



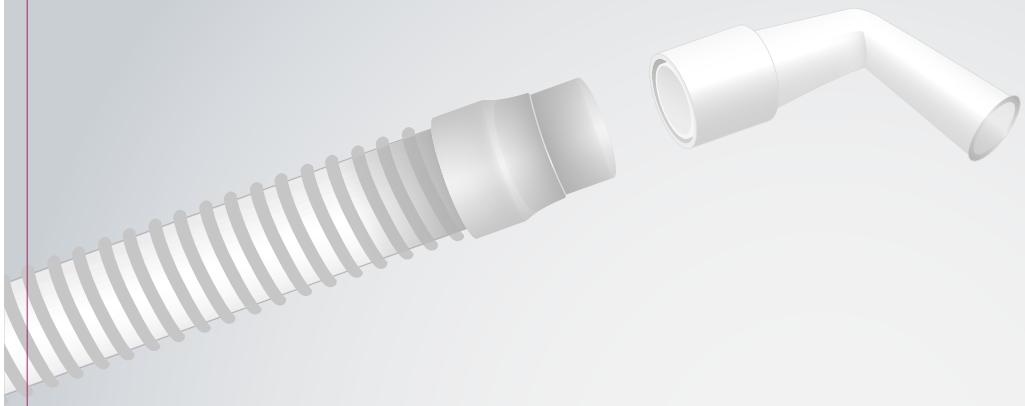
* Please see Aerogen user guide for details.

5.5 mm pressure line



* Please see Aerogen user guide for details.

Note: Adding a HEPA filter at the air inlet helps to prevent inhalation deposits in the exhalation valve during nebulisation.

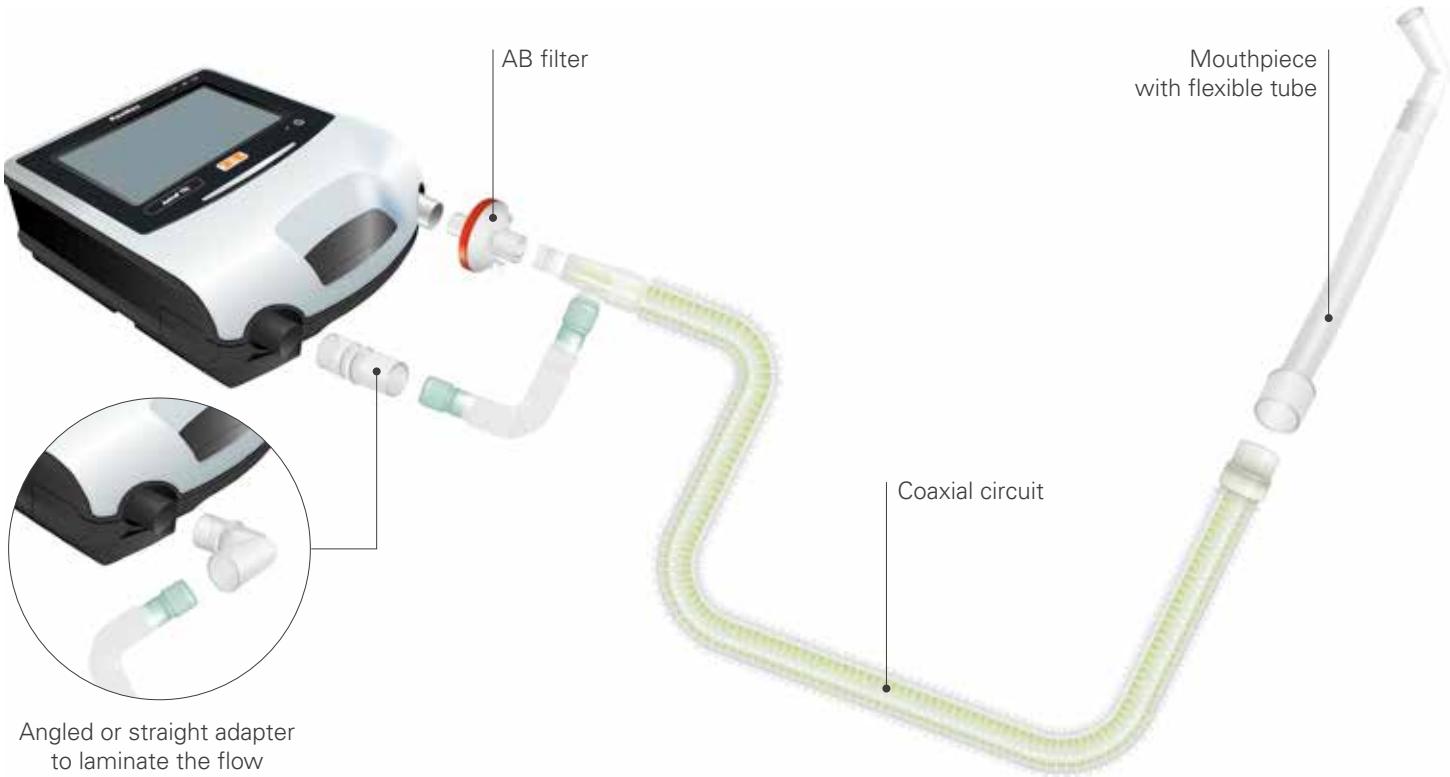


Mouthpiece ventilation

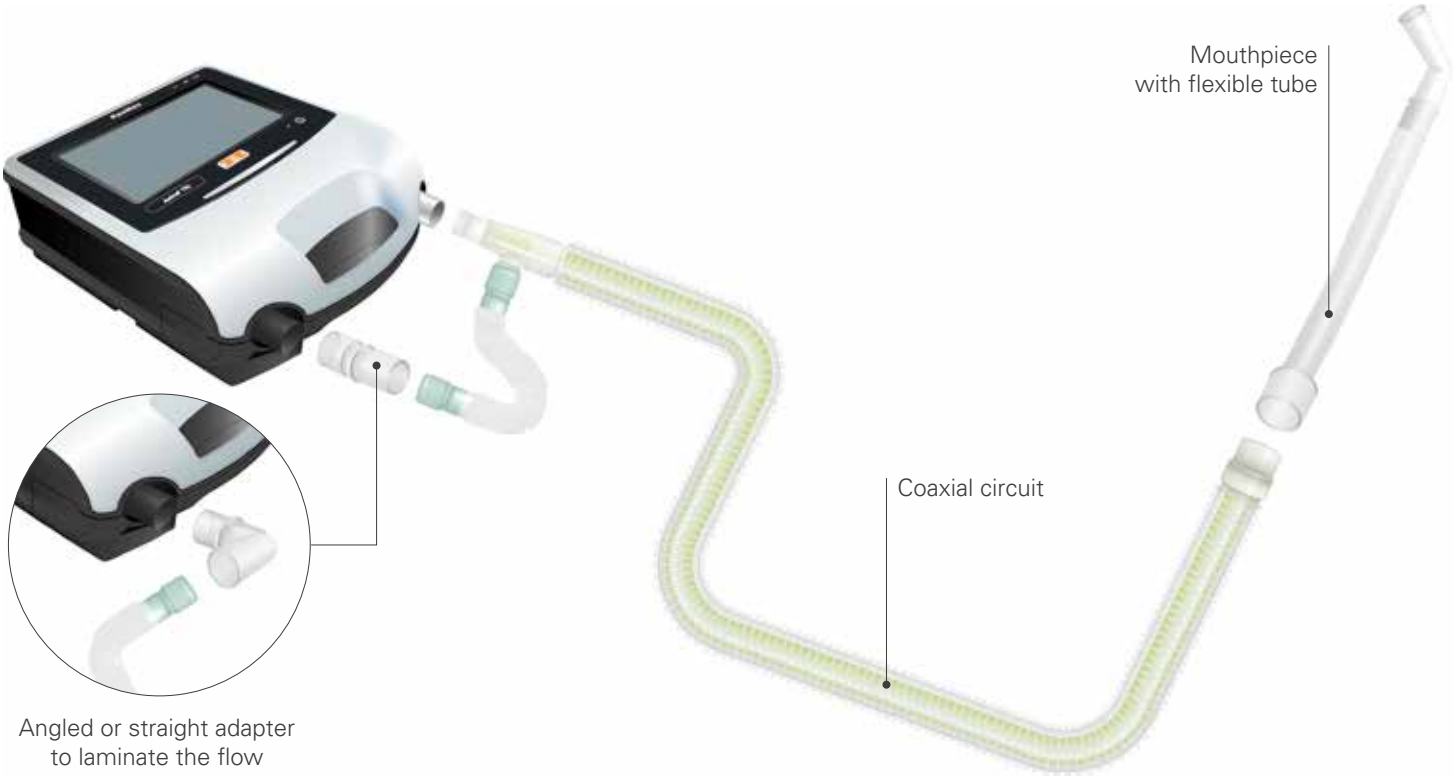


Straight or angled adapter in front of expiratory port

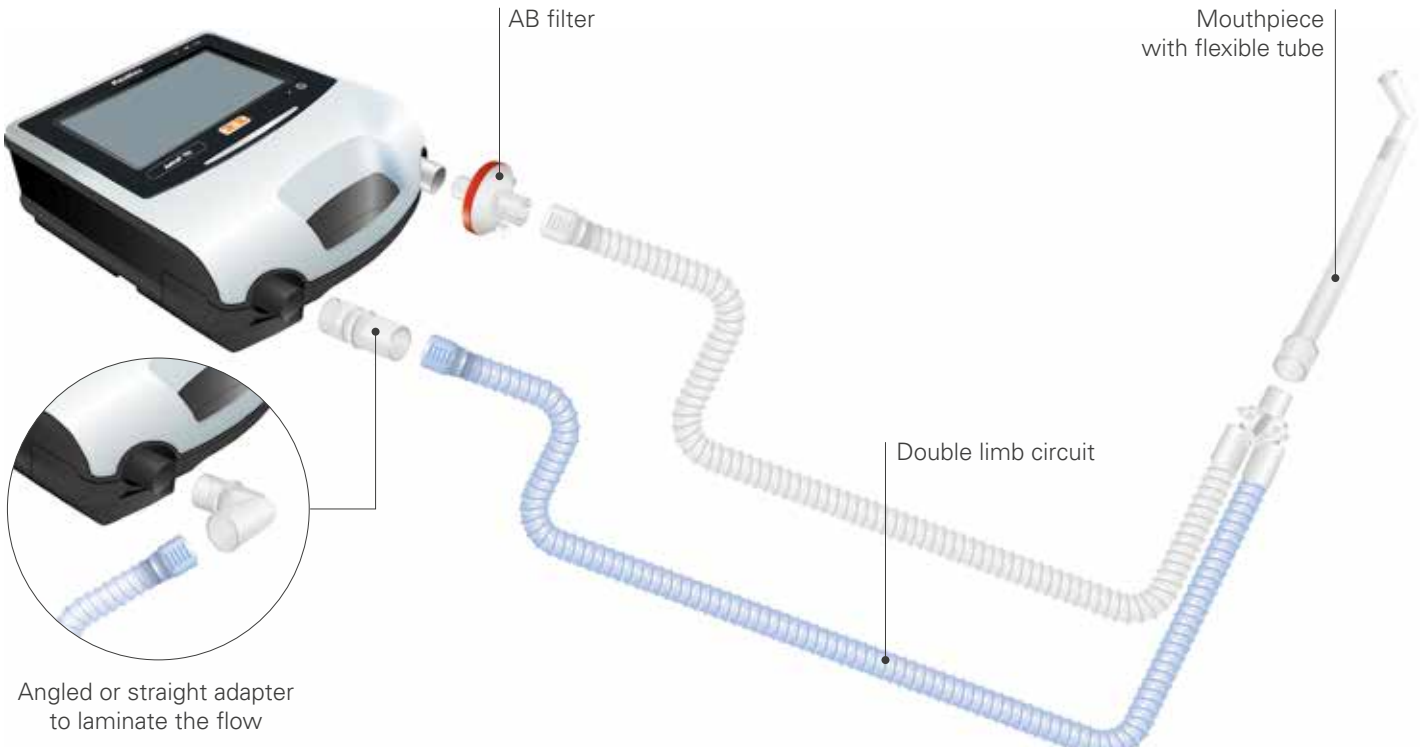
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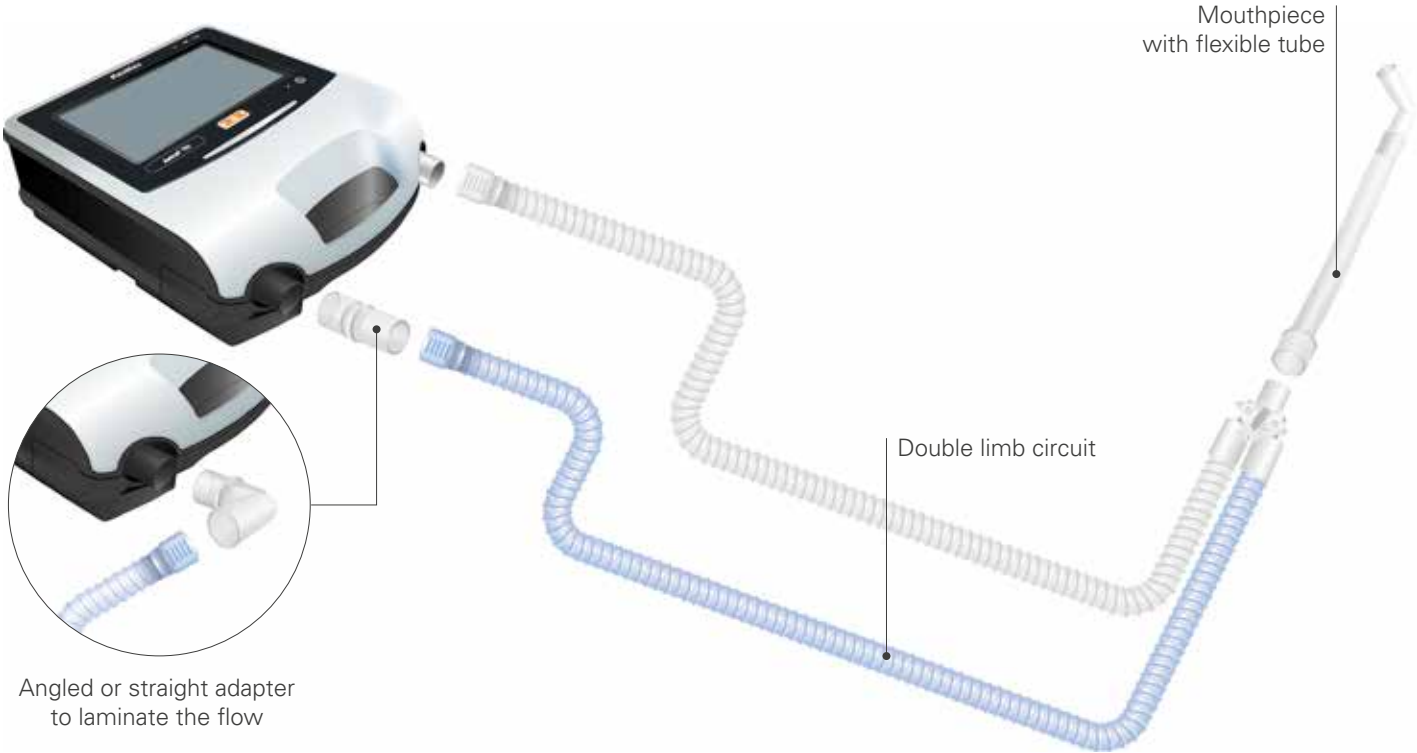
Straight or angled adapter in front of expiratory port

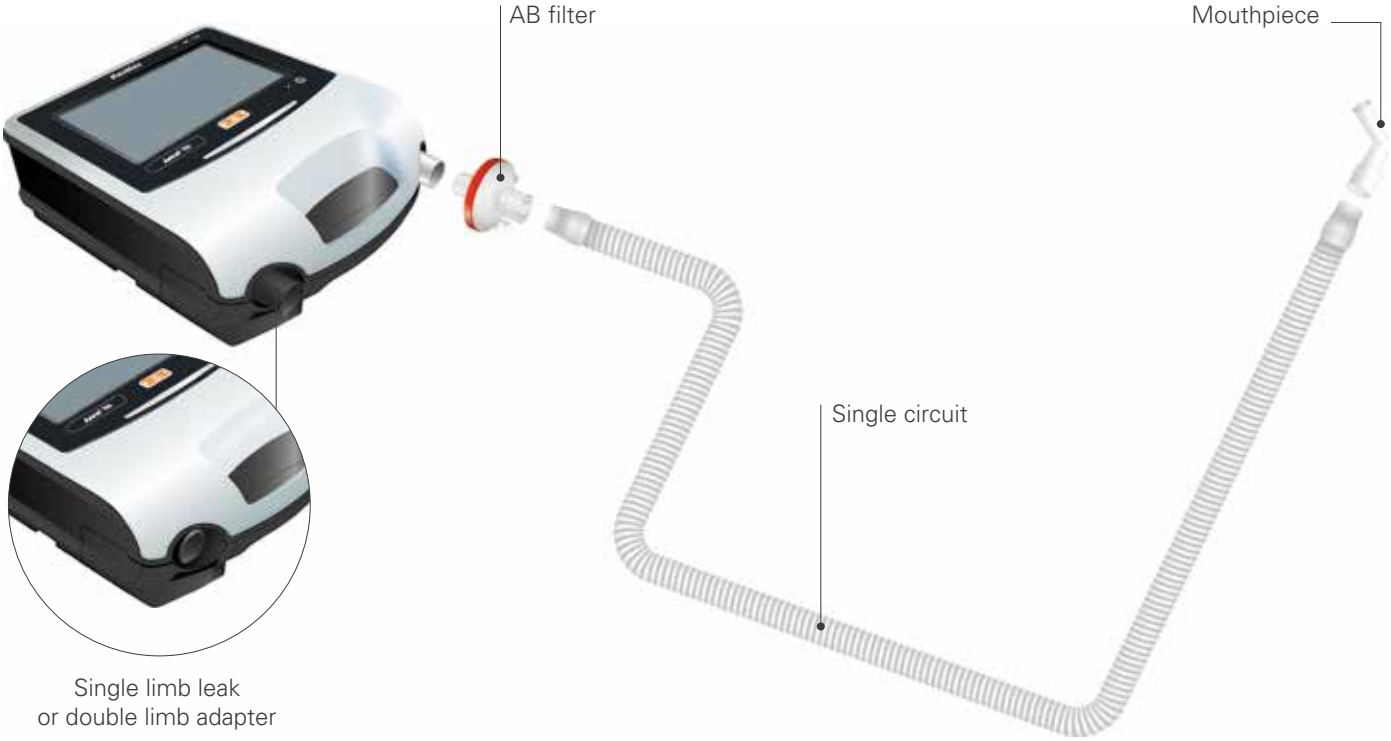


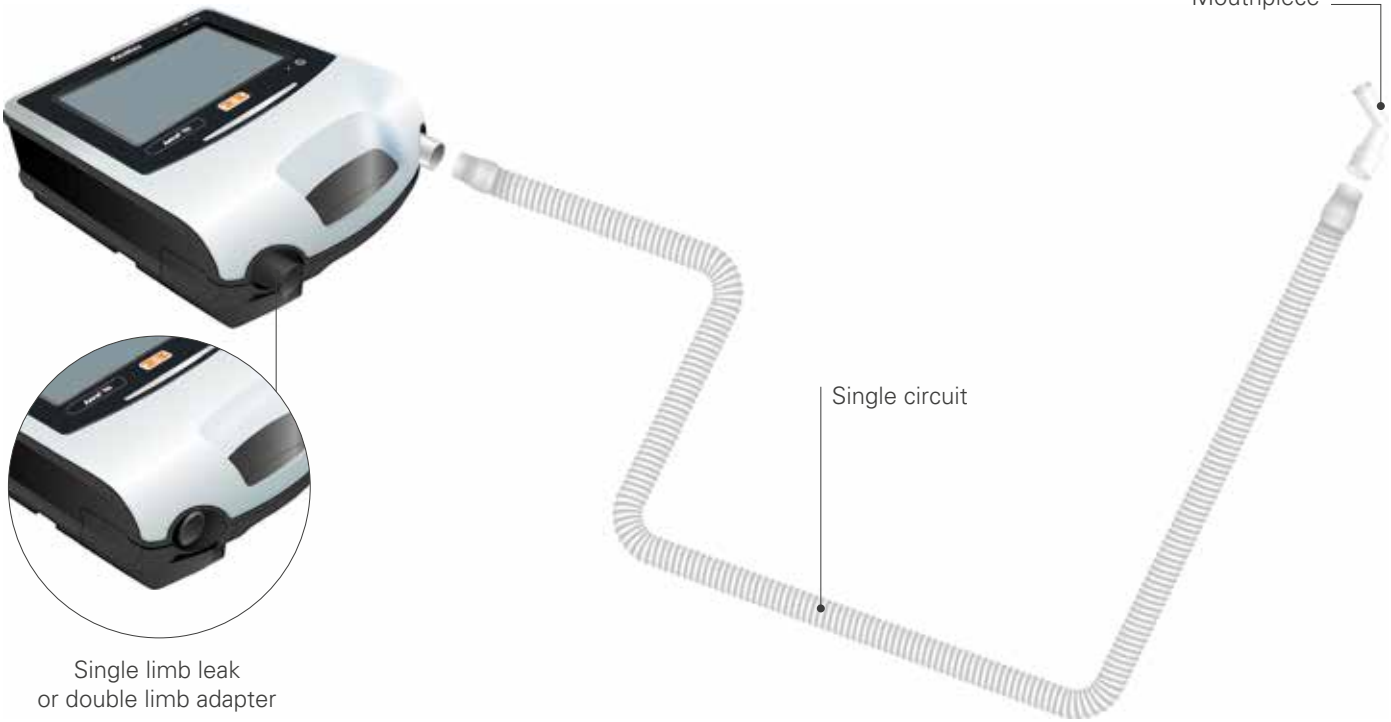
Straight or angled adapter in front of expiratory port

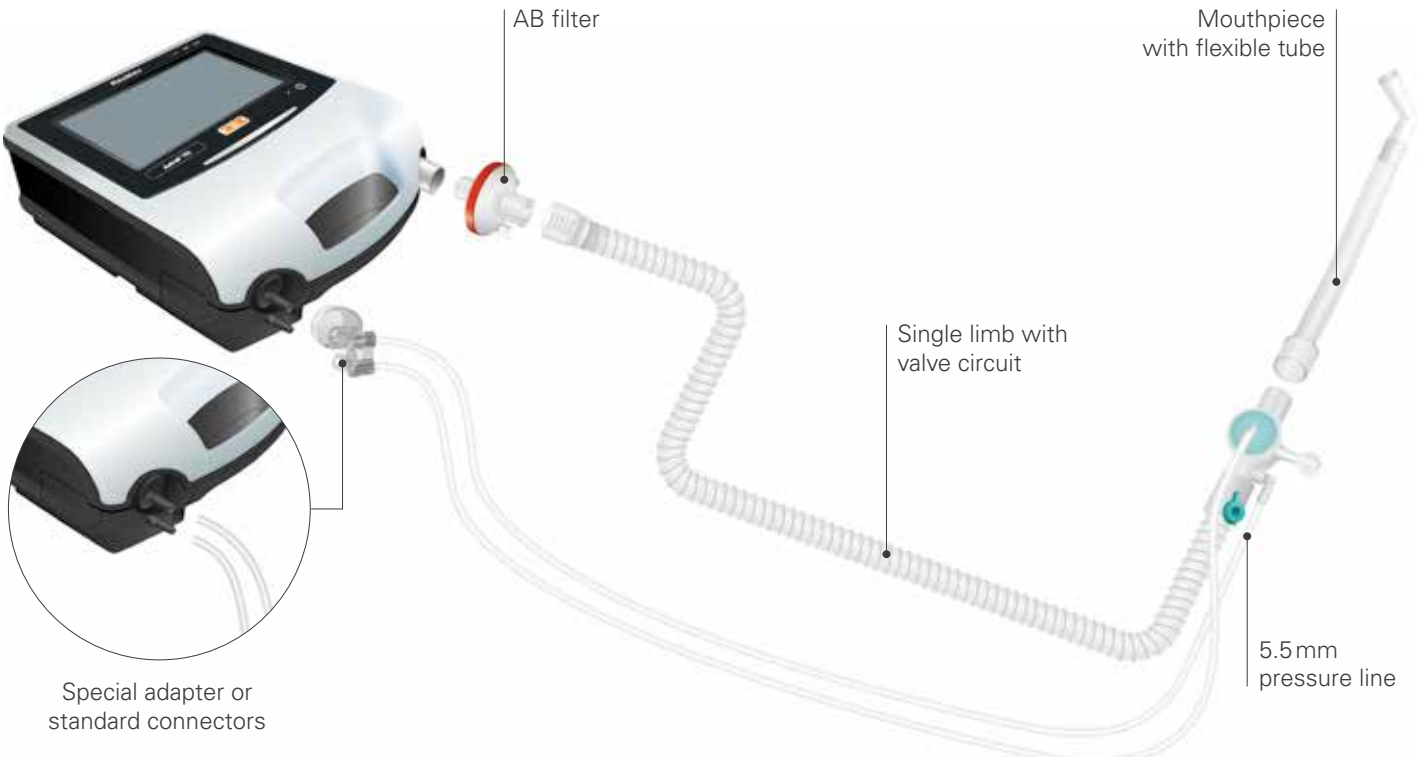


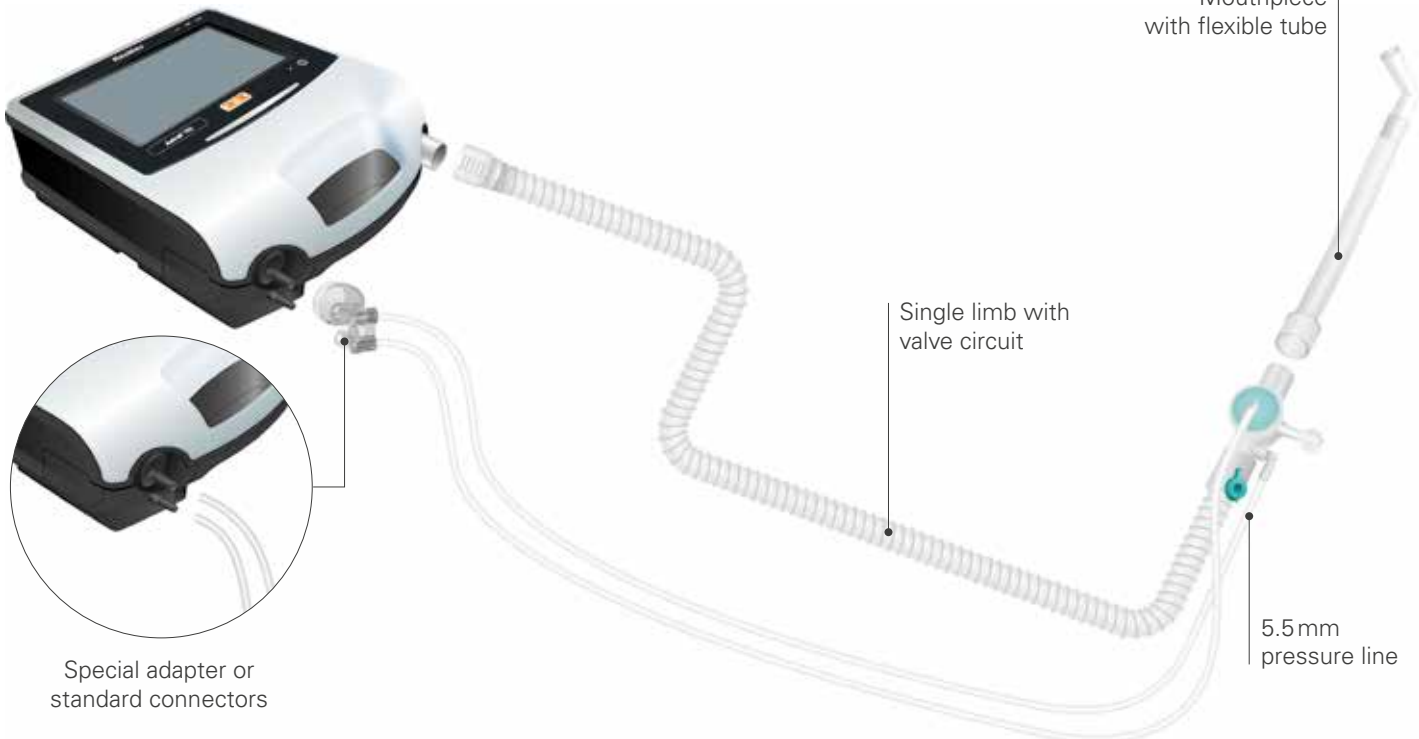
Straight or angled adapter in front of expiratory port




AB filter
Mouthpiece
Single circuit
Single limb leak
or double limb adapter


Mouthpiece
Single circuit
Single limb leak
or double limb adapter


AB filter
Mouthpiece
with flexible tube
Single limb with
valve circuit
5.5 mm
pressure line
Special adapter or
standard connectors


Mouthpiece
with flexible tube
Single limb with
valve circuit
5.5 mm
pressure line
Special adapter or
standard connectors



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